Lab Assignment

International Cybersecurity and Digital Forensics Academy

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

1. Purpose of the Lab Assignment

The purpose of this case study is to investigate simulated harassment email traffic using digital forensic methods. The investigation focuses on a supplied network packet capture and related case materials.
Wireshark is used to identify the web activity connected to the reported harassment message. The investigation also examines the client IP address, MAC address, and available browser or cookie evidence.

2. Lab Report Structure

The report covers the case information, executive conclusion, evidence acquisition, methodology, and investigation findings. It documents the PCAP file, file size, SHA-256 hash, and working copy used during the investigation.
The findings answer the eight investigation questions using packet numbers, timestamps, filters, and screenshots. A timeline is included to show the sequence of relevant network activities.

2.1. Title Page

• Lab Title: Case Study 1: Investigating Harassment Email Traffic with Wireshark

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• Student ID: 2025/FWSD/11320.

• Course Name: SBT-DF204: Computer Forensic Case Study

• Instructor Name: Aminu Idris

• Date of Submission: 04/10/2026

• Version:

2.2. Executive Summary

This case study investigates network traffic related to a reported harassment message sent through a web service. The supplied PCAP is examined using Wireshark and related forensic

analysis commands.
The investigation identifies the client system, web service, relevant HTTP requests, and device information. Cookie or browser evidence is examined to determine whether the device can be linked to an identity in the scenario.

2.3. Lab Objectives

The objectives are to preserve the supplied PCAP and verify its integrity using SHA-256 hashing. The investigation aims to identify the client IP address and service IP address involved in the web activity. It also aims to locate the relevant HTTP POST and correlate it with the reported harassment message.
Another objective is to identify the device and examine scenario provided evidence that may link it to an individual.

2.4. Tools and Resources Used

Wireshark is used to inspect packets, apply display filters, and follow relevant TCP streams.
TShark is used to extract packet numbers, timestamps, IP addresses, MAC addresses, and HTTP information.
SHA-256 hashing is used to verify the integrity of the original and working PCAP files.
The supplied Nitroba PCAP, email material, and Chemistry 109 roster are used as case evidence.
Kali Linux is used as the forensic analysis environment for the command-line investigation.

2.5. Methodology

This individual case study asks you to investigate a simulated harassment-email complaint using supplied email material, a class roster, and a packet capture. Your report must show how you preserved the evidence, found the relevant web activity in Wireshark, and assessed whether the traffic supports attribution to a student in Chemistry 109. Do not assume that an IP address proves identity. Your conclusion must follow the evidence and explain its limits.

Case scenario

Lily Tuckrige, a Chemistry Department teacher, reports repeated harassing emails sent to her personal Yahoo Mail account. Full headers from an earlier message point to an IP address in a shared student residence where an open wireless router is in use. On Monday 21 July, a later message arrived through the web service willselfdestruct.com. The service briefly displayed the message and then reported that it had been destroyed. The university captured traffic at the residence network boundary and asks you to determine whether the available evidence supports attribution to a student in Chemistry 109.

Case materials

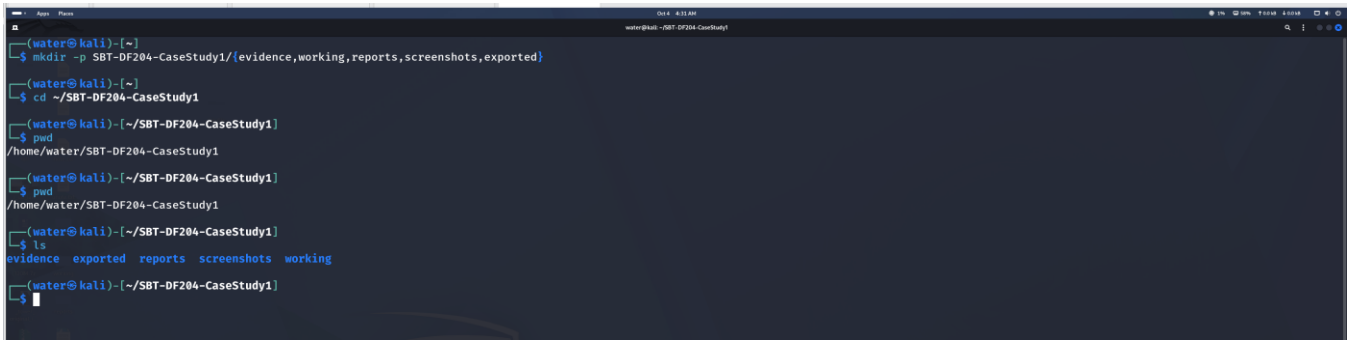
Use the scenario-provided first email, full headers, second email, class roster, and network traffic capture. **Scenario page:** digitalcorpora.org/corpora/scenarios/nitroba-university-harassment-scenario

Traffic capture: digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap

If the source page also provides a published checksum, compare it with the hash you calculate. If it does not, record your calculated SHA-256 value in the report so another examiner can identify the evidence file you analysed.

First is to Create the Case study Folder (SBT-DF204-CaseStudy1)

mkdir -p SBT-DF204-CaseStudy1/{evidence,working,reports,screenshots,exported}



Download the PCAP file from the from the case evidence (nitroba.pcap)

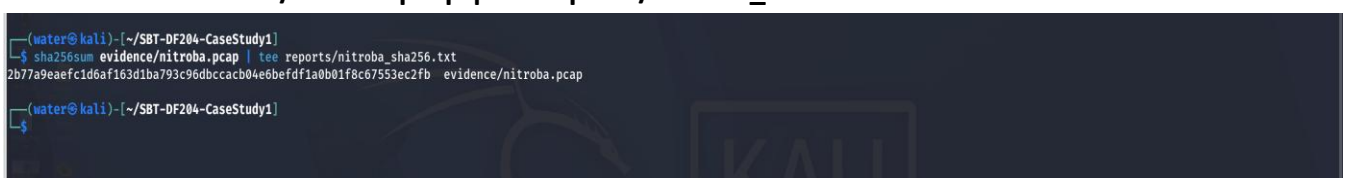
**wget -O evidence/nitroba.pcap **

https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap



Check or calculate the original SHA-256 value, Size and if the evidence is a PCAP capture file

sha256sum evidence/nitroba.pcap | tee reports/nitroba_sha256.txt



```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ stat evidence/nitroba.pcap | tee reports/nitroba_file_stat.txt
  File: evidence/nitroba.pcap
  Size: 56180821        Blocks: 109744     IO Block: 4096   regular file
Device: 8,1     Inode: 1610762     Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   water)   Gid: ( 1000/   water)
Access: 2026-10-04 04:37:13.238838061 -0400
Modify: 2020-11-21 22:36:00.000000000 -0500
Change: 2026-10-04 04:34:28.130120110 -0400
 Birth: 2026-10-04 04:34:15.162068264 -0400

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ file evidence/nitroba.pcap | tee reports/nitroba_file_type.txt
evidence/nitroba.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 4096)

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$
```

For Forensic working copy: Download the scenario traffic capture and work only from your copied evidence file. Record the original file name, source link, file size, download date, and SHA-256 value

Making a forensic working and copy and also checking the SHA-256
cp --preserve=all evidence/nitroba.pcap working/nitroba_working.pcap
sha256sum evidence/nitroba.pcap working/nitroba_working.pcap

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ cp --preserve=all evidence/nitroba.pcap working/nitroba_working.pcap

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ sha256sum evidence/nitroba.pcap working/nitroba_working.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  evidence/nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  working/nitroba_working.pcap

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ sha256sum evidence/nitroba.pcap working/nitroba_working.pcap \
| tee reports/working_copy_hash_verification.txt
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  evidence/nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  working/nitroba_working.pcap
```

Both the Original evidence and working copy SHA-256 value are the same

Confirming the PCAP details/Information before working on Wireshark
capinfos working/nitroba_working.pcap

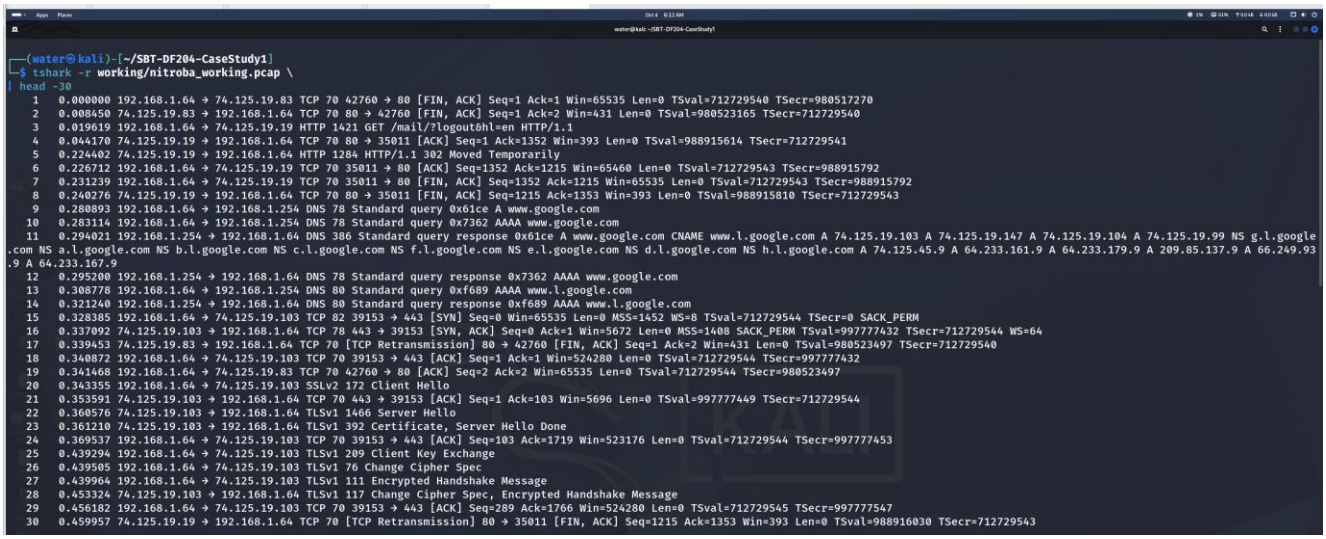
```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ capinfos working/nitroba_working.pcap
File name:           working/nitroba_working.pcap
File type:           Wireshark/tcpdump/... - pcap
File encapsulation:  Ethernet
File timestamp precision:  microseconds (6)
Packet size limit:   file hdr: 4096 bytes
Number of packets:   94 k
File size:           56 MB
Data size:           54 MB
Capture duration:    15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time:   2008-07-22 02:13:47.046029
Data byte rate:      3,468 bytes/s
Data bit rate:       27 kbps
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
SHA256:              2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA1:                65656392412add15f93f8585197a8998aaeb50a1
Strict time order:   True
Number of interfaces in file: 1
Interface #0 info:
                     Encapsulation = Ethernet (1 - ether)
                     Capture length = 4096
                     Time precision = microseconds (6)
                     Time ticks per second = 1000000
                     Number of stat entries = 0
                     Number of packets = 94410
```

Wireshark analysis: Use display filters, packet inspection, and Follow TCP Stream where appropriate. Preserve packet numbers and exact timestamps for every material finding.

Using Tshark to examine the nitroba_working.pcap before using Wireshark

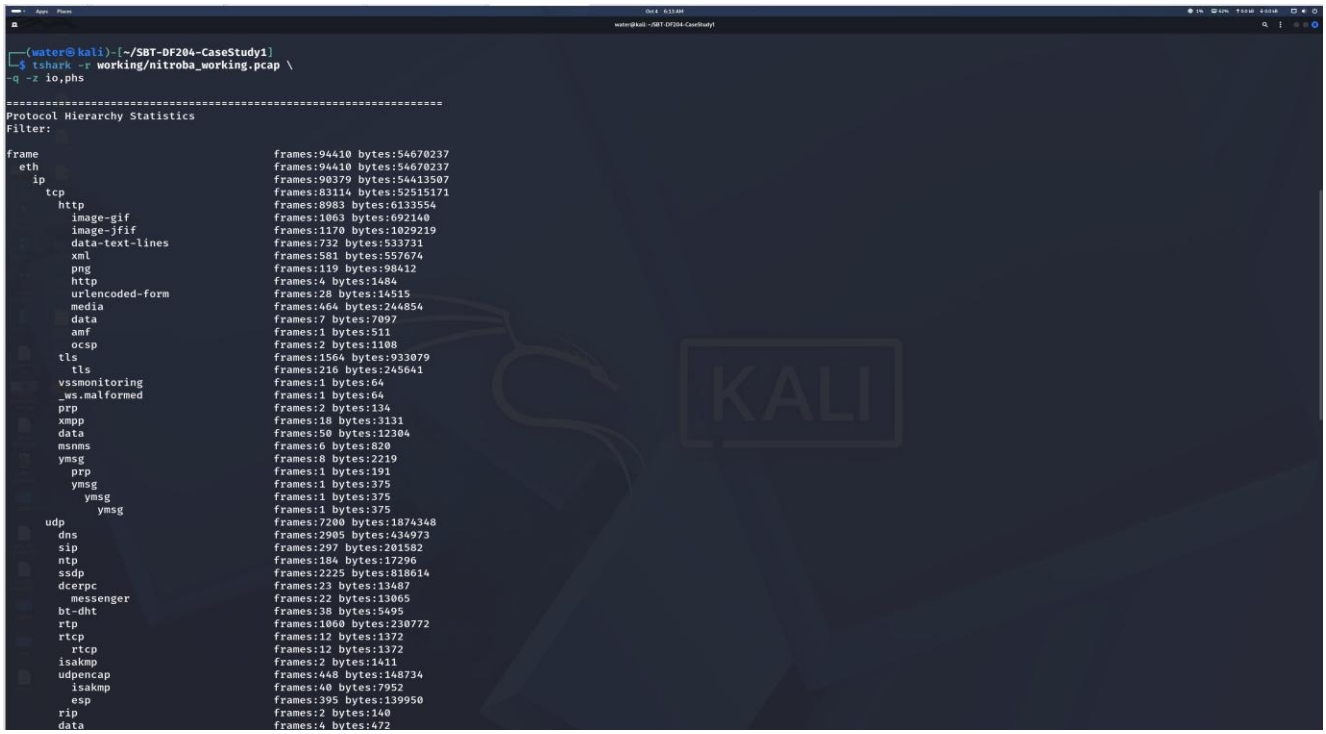
Opening the PCAP in Tshark

tshark -r working/nitroba_working.pcap \ | head -30



```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ tshark -r working/nitroba_working.pcap \
| head -30
    1   0.000000 192.168.1.64 → 74.125.19.83 TCP 70 42760 → 80 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0 TSval=712729540 TSecr=980517270
    2   0.008450 74.125.19.83 → 192.168.1.64 TCP 70 80 → 42760 [FIN, ACK] Seq=1 Ack=2 Win=431 Len=0 TSval=980523165 TSecr=712729540
    3   0.019619 192.168.1.64 → 74.125.19.19 HTTP 1421 GET /mail/?logout&hl=en HTTP/1.1
    4   0.044170 74.125.19.19 → 192.168.1.64 TCP 70 80 → 35011 [ACK] Seq=1 Ack=1352 Win=393 Len=0 TSval=988915614 TSecr=712729541
    5   0.224402 74.125.19.19 → 192.168.1.64 HTTP 1284 HTTP/1.1 302 Moved Temporarily
    6   0.226712 192.168.1.64 → 74.125.19.19 TCP 70 35011 → 80 [ACK] Seq=1352 Ack=1215 Win=65460 Len=0 TSval=712729543 TSecr=988915792
    7   0.231239 192.168.1.64 → 74.125.19.19 TCP 70 35011 → 80 [FIN, ACK] Seq=1352 Ack=1215 Win=65535 Len=0 TSval=712729543 TSecr=988915792
    8   0.240276 74.125.19.19 → 192.168.1.64 TCP 70 80 → 35011 [FIN, ACK] Seq=1215 Ack=1353 Win=393 Len=0 TSval=988915810 TSecr=712729543
    9   0.280893 192.168.1.64 → 192.168.1.254 DNS 78 Standard query 0x61ce A www.google.com
   10   0.283114 192.168.1.64 → 192.168.1.254 DNS 78 Standard query 0x7362 AAAA www.google.com
   11   0.294021 192.168.1.254 → 192.168.1.64 DNS 386 Standard query response 0x61ce A www.google.com CNAME www.l.google.com A 74.125.19.103 A 74.125.19.147 A 74.125.19.104 A 74.125.19.99 NS g.l.google
.com NS a.l.google.com NS b.l.google.com NS c.l.google.com NS f.l.google.com NS e.l.google.com NS d.l.google.com NS h.l.google.com A 74.125.45.9 A 64.233.161.9 A 64.233.179.9 A 209.85.137.9 A 66.249.93
.9 A 64.233.167.9
   12   0.295200 192.168.1.254 → 192.168.1.64 DNS 78 Standard query response 0x7362 AAAA www.google.com
   13   0.308778 192.168.1.64 → 192.168.1.254 DNS 80 Standard query 0xf689 AAAA www.l.google.com
   14   0.321240 192.168.1.254 → 192.168.1.64 DNS 80 Standard query response 0xf689 AAAA www.l.google.com
   15   0.328385 192.168.1.64 → 74.125.19.103 TCP 82 39153 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1452 WS=8 TSval=712729544 TSecr=0 SACK_PERM
   16   0.337092 74.125.19.103 → 192.168.1.64 TCP 78 443 → 39153 [SYN, ACK] Seq=0 Ack=1 Win=5672 Len=0 MSS=1408 SACK_PERM TSval=997777432 TSecr=712729544 WS=64
   17   0.339453 74.125.19.83 → 192.168.1.64 TCP 70 [TCP Retransmission] 80 → 42760 [FIN, ACK] Seq=1 Ack=2 Win=431 Len=0 TSval=980523497 TSecr=712729540
   18   0.340872 192.168.1.64 → 74.125.19.103 TCP 70 39153 → 443 [ACK] Seq=1 Ack=1 Win=524280 Len=0 TSval=712729544 TSecr=997777432
   19   0.341468 192.168.1.64 → 74.125.19.83 TCP 70 42760 → 80 [ACK] Seq=2 Ack=2 Win=65535 Len=0 TSval=712729544 TSecr=980523497
   20   0.343355 192.168.1.64 → 74.125.19.103 SSLv2 172 Client Hello
   21   0.353591 74.125.19.103 → 192.168.1.64 TCP 70 443 → 39153 [ACK] Seq=1 Ack=103 Win=5696 Len=0 TSval=997777449 TSecr=712729544
   22   0.360576 74.125.19.103 → 192.168.1.64 TLSv1 1466 Server Hello
   23   0.361210 74.125.19.103 → 192.168.1.64 TLSv1 392 Certificate, Server Hello Done
   24   0.369537 192.168.1.64 → 74.125.19.103 TCP 70 39153 → 443 [ACK] Seq=103 Ack=1719 Win=523176 Len=0 TSval=712729544 TSecr=997777453
   25   0.439294 192.168.1.64 → 74.125.19.103 TLSv1 209 Client Key Exchange
   26   0.439505 192.168.1.64 → 74.125.19.103 TLSv1 76 Change Cipher Spec
   27   0.439964 192.168.1.64 → 74.125.19.103 TLSv1 111 Encrypted Handshake Message
   28   0.453324 74.125.19.103 → 192.168.1.64 TLSv1 117 Change Cipher Spec, Encrypted Handshake Message
   29   0.456182 192.168.1.64 → 74.125.19.103 TCP 70 39153 → 443 [ACK] Seq=289 Ack=1766 Win=524280 Len=0 TSval=712729545 TSecr=997777547
   30   0.459957 74.125.19.19 → 192.168.1.64 TCP 70 [TCP Retransmission] 80 → 35011 [FIN, ACK] Seq=1215 Ack=1353 Win=393 Len=0 TSval=988916030 TSecr=712729543
```

tshark -r working/nitroba_working.pcap \ -q -z io,phs



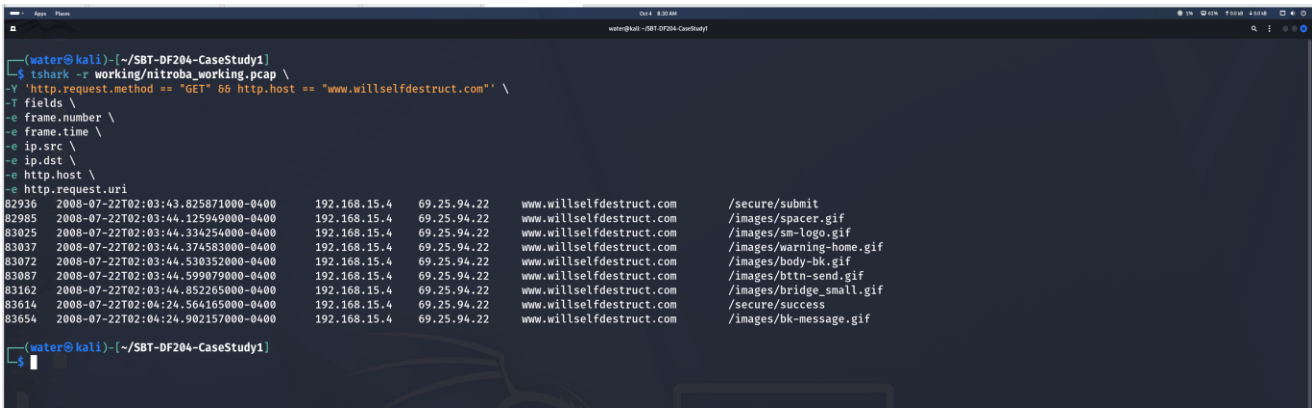
```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ tshark -r working/nitroba_working.pcap \
-q -z io,phs

===================================================================
Protocol Hierarchy Statistics
Filter:

frame                                    frames:94410 bytes:54670237
  eth                                    frames:94410 bytes:54670237
    ip                                   frames:90379 bytes:54413507
      tcp                                frames:83114 bytes:52515171
        http                             frames:8983 bytes:6133554
          image-gif                      frames:1063 bytes:692140
          image-jfif                     frames:1170 bytes:1029219
          data-text-lines                frames:732 bytes:533731
          xml                            frames:581 bytes:557674
          png                            frames:119 bytes:98412
          http                           frames:4 bytes:1484
          urlencoded-form                frames:28 bytes:14515
          media                          frames:464 bytes:244854
          data                           frames:7 bytes:7097
          amf                            frames:1 bytes:511
          ocsp                           frames:2 bytes:1108
        tls                              frames:1564 bytes:933079
          tls                            frames:216 bytes:245641
        vssmonitoring                    frames:1 bytes:64
        _ws.malformed                    frames:1 bytes:64
        prp                              frames:2 bytes:134
        xmpp                             frames:18 bytes:3131
        data                             frames:50 bytes:12304
        msnms                            frames:6 bytes:820
        ymsg                             frames:8 bytes:2219
          prp                            frames:1 bytes:191
          ymsg                           frames:1 bytes:375
            ymsg                         frames:1 bytes:375
              ymsg                       frames:1 bytes:375
      udp                                frames:7200 bytes:1874348
        dns                              frames:2905 bytes:434973
        sip                              frames:297 bytes:201582
        ntp                              frames:184 bytes:17296
        ssdp                             frames:2225 bytes:818614
        dcerpc                           frames:23 bytes:13487
          messenger                      frames:22 bytes:13065
        bt-dht                           frames:38 bytes:5495
        rtp                              frames:1060 bytes:230772
        rtcp                             frames:12 bytes:1372
          rtcp                           frames:12 bytes:1372
        isakmp                           frames:2 bytes:1411
        udpencap                         frames:448 bytes:148734
          isakmp                         frames:40 bytes:7952
          esp                            frames:395 bytes:139950
        rip                              frames:2 bytes:140
        data                             frames:4 bytes:472
```

Finding the willselfdestruct.com traffic on the Tshark

tshark -r working/nitroba_working.pcap \ -Y 'http.request.method == "GET" && http.host == "www.willselfdestruct.com"' \ -T fields \ -e frame.number \ -e frame.time \ -e ip.src \ -e ip.dst \ -e http.host \ -e http.request.uri

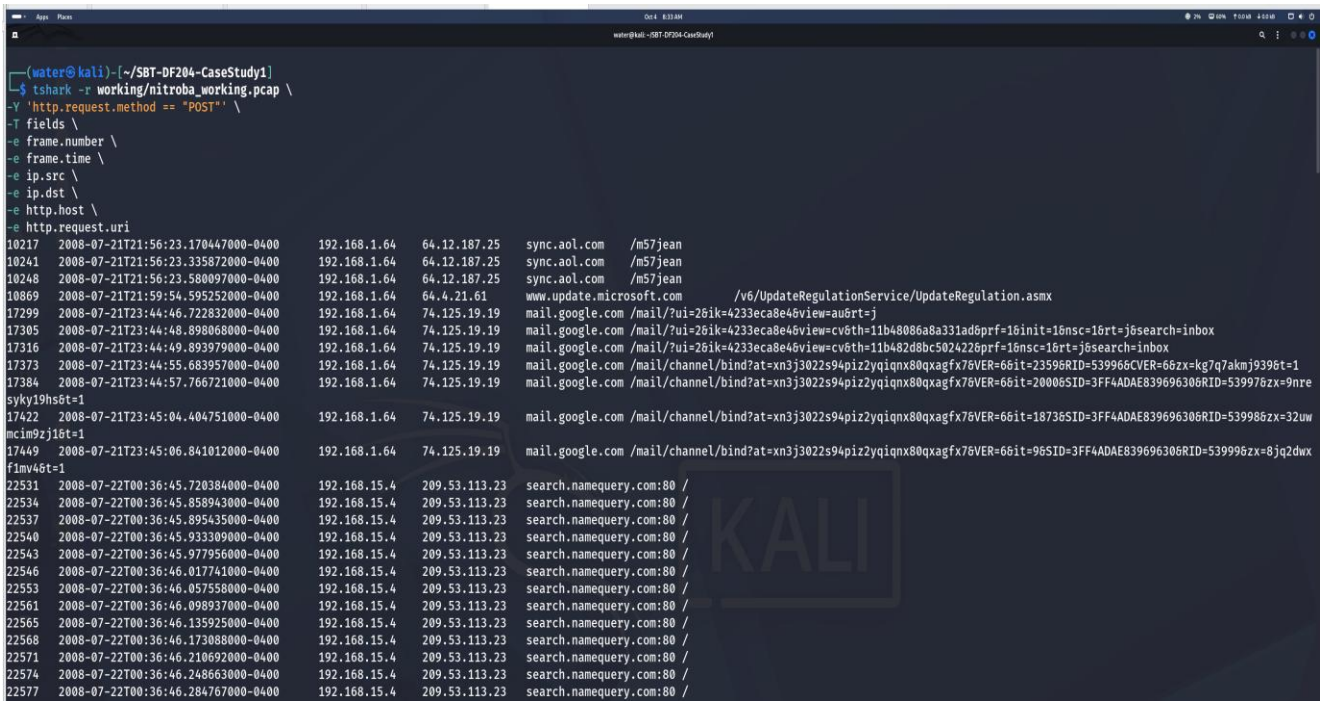


```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ tshark -r working/nitroba_working.pcap \
-Y 'http.request.method == "GET" && http.host == "www.willselfdestruct.com"' \
-T fields \
-e frame.number \
-e frame.time \
-e ip.src \
-e ip.dst \
-e http.host \
-e http.request.uri
82936   2008-07-22T02:03:43.825871000-0400      192.168.15.4    69.25.94.22     www.willselfdestruct.com        /secure/submit
82985   2008-07-22T02:03:44.125949000-0400      192.168.15.4    69.25.94.22     www.willselfdestruct.com        /images/spacer.gif
83025   2008-07-22T02:03:44.334254000-0400      192.168.15.4    69.25.94.22     www.willselfdestruct.com        /images/sm-logo.gif
83037   2008-07-22T02:03:44.374583000-0400      192.168.15.4    69.25.94.22     www.willselfdestruct.com        /images/warning-home.gif
83072   2008-07-22T02:03:44.530352000-0400      192.168.15.4    69.25.94.22     www.willselfdestruct.com        /images/body-bk.gif
83087   2008-07-22T02:03:44.599079000-0400      192.168.15.4    69.25.94.22     www.willselfdestruct.com        /images/bttn-send.gif
83162   2008-07-22T02:03:44.852265000-0400      192.168.15.4    69.25.94.22     www.willselfdestruct.com        /images/bridge_small.gif
83614   2008-07-22T02:04:24.564165000-0400      192.168.15.4    69.25.94.22     www.willselfdestruct.com        /secure/success
83654   2008-07-22T02:04:24.902157000-0400      192.168.15.4    69.25.94.22     www.willselfdestruct.com        /images/bk-message.gif

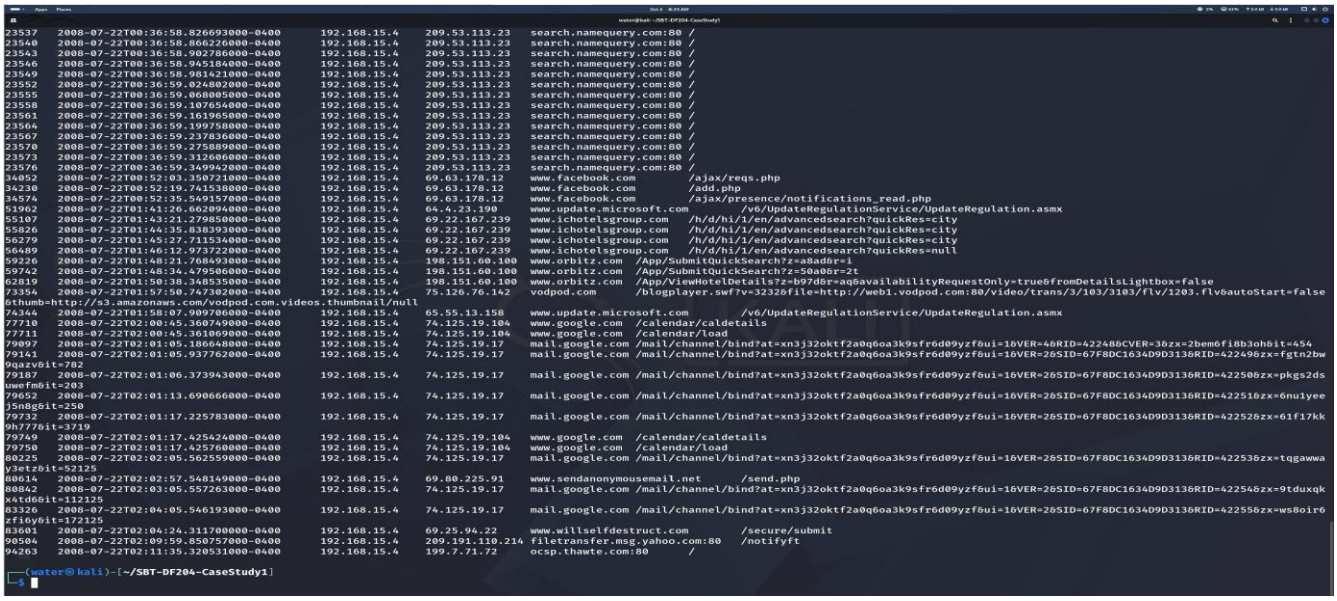
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$
```

Finding the HTTP POST with Tshark

tshark -r working/nitroba_working.pcap \ -Y 'http.request.method == "POST"' \ -T fields \ -e frame.number \ -e frame.time \ -e ip.src \ -e ip.dst \ -e http.host \ -e http.request.uri



```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ tshark -r working/nitroba_working.pcap \
-Y 'http.request.method == "POST"' \
-T fields \
-e frame.number \
-e frame.time \
-e ip.src \
-e ip.dst \
-e http.host \
-e http.request.uri
10217   2008-07-21T21:56:23.170447000-0400      192.168.1.64    64.12.187.25    sync.aol.com    /m57jean
10241   2008-07-21T21:56:23.335872000-0400      192.168.1.64    64.12.187.25    sync.aol.com    /m57jean
10248   2008-07-21T21:56:23.580097000-0400      192.168.1.64    64.12.187.25    sync.aol.com    /m57jean
10869   2008-07-21T21:59:54.595252000-0400      192.168.1.64    64.4.21.61      www.update.microsoft.com        /v6/UpdateRegulationService/UpdateRegulation.asmx
17299   2008-07-21T23:44:46.722832000-0400      192.168.1.64    74.125.19.19    mail.google.com /mail/?ui=2&ik=4233eca8e4&view=au&rt=j
17305   2008-07-21T23:44:48.898068000-0400      192.168.1.64    74.125.19.19    mail.google.com /mail/?ui=2&ik=4233eca8e4&view=cv&th=11b48086a8a331ad&prf=1&init=1&nsc=1&rt=j&search=inbox
17316   2008-07-21T23:44:49.893979000-0400      192.168.1.64    74.125.19.19    mail.google.com /mail/?ui=2&ik=4233eca8e4&view=cv&th=11b482d8bc502422&prf=1&nsc=1&rt=j&search=inbox
17373   2008-07-21T23:44:55.683957000-0400      192.168.1.64    74.125.19.19    mail.google.com /mail/channel/bind?at=xn3j3022s94piz2yqiqnx80qxagfx7&VER=6&it=23596&RID=53996&CVER=6&zx=kg7q7akmj939&t=1
17384   2008-07-21T23:44:57.766721000-0400      192.168.1.64    74.125.19.19    mail.google.com /mail/channel/bind?at=xn3j3022s94piz2yqiqnx80qxagfx7&VER=6&it=20006&SID=3FF4ADAE83969630&RID=53997&zx=9nre
syky19hs&t=1
17422   2008-07-21T23:45:04.404751000-0400      192.168.1.64    74.125.19.19    mail.google.com /mail/channel/bind?at=xn3j3022s94piz2yqiqnx80qxagfx7&VER=6&it=18736&SID=3FF4ADAE83969630&RID=53998&zx=32uw
mcim9zj1&t=1
17449   2008-07-21T23:45:06.841012000-0400      192.168.1.64    74.125.19.19    mail.google.com /mail/channel/bind?at=xn3j3022s94piz2yqiqnx80qxagfx7&VER=6&it=96&SID=3FF4ADAE83969630&RID=53999&zx=8jq2dwx
f1mv4&t=1
22531   2008-07-22T00:36:45.720384000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
22534   2008-07-22T00:36:45.858943000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
22537   2008-07-22T00:36:45.895435000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
22540   2008-07-22T00:36:45.933309000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
22543   2008-07-22T00:36:45.977956000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
22546   2008-07-22T00:36:46.017741000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
22553   2008-07-22T00:36:46.057558000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
22561   2008-07-22T00:36:46.098937000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
22565   2008-07-22T00:36:46.135925000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
22568   2008-07-22T00:36:46.173088000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
22571   2008-07-22T00:36:46.210692000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
22574   2008-07-22T00:36:46.248663000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
22577   2008-07-22T00:36:46.284767000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
```



```
23537   2008-07-22T00:36:58.826693000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
23540   2008-07-22T00:36:58.866226000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
23543   2008-07-22T00:36:58.902786000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
23546   2008-07-22T00:36:58.945184000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
23549   2008-07-22T00:36:58.981421000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
23552   2008-07-22T00:36:59.024802000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
23555   2008-07-22T00:36:59.068005000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
23558   2008-07-22T00:36:59.107654000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
23561   2008-07-22T00:36:59.161965000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
23564   2008-07-22T00:36:59.199758000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
23567   2008-07-22T00:36:59.237836000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
23570   2008-07-22T00:36:59.275889000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
23573   2008-07-22T00:36:59.312606000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
23576   2008-07-22T00:36:59.349942000-0400      192.168.15.4    209.53.113.23   search.namequery.com:80 /
34052   2008-07-22T00:52:03.350721000-0400      192.168.15.4    69.63.178.12    www.facebook.com        /ajax/reqs.php
34230   2008-07-22T00:52:19.741538000-0400      192.168.15.4    69.63.178.12    www.facebook.com        /add.php
34574   2008-07-22T00:52:35.549157000-0400      192.168.15.4    69.63.178.12    www.facebook.com        /ajax/presence/notifications_read.php
51962   2008-07-22T01:41:26.662094000-0400      192.168.15.4    64.4.23.190     www.update.microsoft.com        /v6/UpdateRegulationService/UpdateRegulation.asmx
55107   2008-07-22T01:43:21.279850000-0400      192.168.15.4    69.22.167.239   www.ichotelsgroup.com   /h/d/hi/1/en/advancedsearch?quickRes=city
55826   2008-07-22T01:44:35.838393000-0400      192.168.15.4    69.22.167.239   www.ichotelsgroup.com   /h/d/hi/1/en/advancedsearch?quickRes=city
56279   2008-07-22T01:45:27.711534000-0400      192.168.15.4    69.22.167.239   www.ichotelsgroup.com   /h/d/hi/1/en/advancedsearch?quickRes=city
56489   2008-07-22T01:46:12.973722000-0400      192.168.15.4    69.22.167.239   www.ichotelsgroup.com   /h/d/hi/1/en/advancedsearch?quickRes=null
59226   2008-07-22T01:48:21.768693000-0400      192.168.15.4    198.151.60.100  www.orbitz.com  /App/SubmitQuickSearch?z=a8ad0r=1
59742   2008-07-22T01:48:34.479506000-0400      192.168.15.4    198.151.60.100  www.orbitz.com  /App/SubmitQuickSearch?z=50a00r=2t
62819   2008-07-22T01:50:38.348535000-0400      192.168.15.4    198.151.60.100  www.orbitz.com  /App/ViewHotelDetails?z=b97d0r=aq0availabilityRequestOnly=true0fromDetailsLightbox=false
73354   2008-07-22T01:57:50.747302000-0400      192.168.15.4    75.126.76.142   vodpod.com      /blogplayer.swf?v=32328file=http://web1.vodpod.com:80/video/trans/3/103/3103/flv/1203.flv0autoStart=false
0thumb=http://s3.amazonaws.com/vodpod.com.videos.thumbnail/null
74344   2008-07-22T01:58:07.909706000-0400      192.168.15.4    65.55.13.158    www.update.microsoft.com        /v6/UpdateRegulationService/UpdateRegulation.asmx
77710   2008-07-22T02:00:45.360749000-0400      192.168.15.4    74.125.19.104   www.google.com  /calendar/caldetails
77711   2008-07-22T02:00:45.361069000-0400      192.168.15.4    74.125.19.104   www.google.com  /calendar/load
79097   2008-07-22T02:01:05.186648000-0400      192.168.15.4    74.125.19.17    mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf8ui=10VER=46RID=42248bCVER=36zx=2bem6fi8b3oh6it=454
79141   2008-07-22T02:01:05.937762000-0400      192.168.15.4    74.125.19.17    mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf8ui=10VER=26SID=67F8DC1634D9D3136RID=422496zx=fgtn2bw
9qazv6it=782
79187   2008-07-22T02:01:06.373943000-0400      192.168.15.4    74.125.19.17    mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf8ui=10VER=26SID=67F8DC1634D9D3136RID=422506zx=pkgs2ds
uwef6it=203
79652   2008-07-22T02:01:13.690666000-0400      192.168.15.4    74.125.19.17    mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf8ui=10VER=26SID=67F8DC1634D9D3136RID=422516zx=6nu1yee
j5n8g6it=250
79732   2008-07-22T02:01:17.225783000-0400      192.168.15.4    74.125.19.17    mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf8ui=10VER=26SID=67F8DC1634D9D3136RID=422526zx=61f17kk
9h776it=3719
79749   2008-07-22T02:01:17.425424000-0400      192.168.15.4    74.125.19.104   www.google.com  /calendar/caldetails
79750   2008-07-22T02:01:17.425760000-0400      192.168.15.4    74.125.19.104   www.google.com  /calendar/load
80225   2008-07-22T02:02:05.562559000-0400      192.168.15.4    74.125.19.17    mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf8ui=10VER=26SID=67F8DC1634D9D3136RID=422536zx=tqgawwa
y3etz6it=52125
80614   2008-07-22T02:02:57.548149000-0400      192.168.15.4    69.80.225.91    www.sendanonymousemail.net      /send.php
80842   2008-07-22T02:03:05.557263000-0400      192.168.15.4    74.125.19.17    mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf8ui=10VER=26SID=67F8DC1634D9D3136RID=422546zx=9tduxqk
x4td66it=112125
83326   2008-07-22T02:04:05.546193000-0400      192.168.15.4    74.125.19.17    mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf8ui=10VER=26SID=67F8DC1634D9D3136RID=422556zx=ws8oir6
zfi6y6it=172125
83601   2008-07-22T02:04:24.311700000-0400      192.168.15.4    69.25.94.22     www.willselfdestruct.com        /secure/submit
90504   2008-07-22T02:09:59.850757000-0400      192.168.15.4    209.191.110.214 filetransfer.msg.yahoo.com:80   /notifyft
94263   2008-07-22T02:11:35.320531000-0400      192.168.15.4    199.7.71.72     ocsp.thawte.com:80      /

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$
```

Find the client's MAC address with Tshark

tshark -r working/nitroba_working.pcap \-Y 'ip.addr == 192.168.15.4' \-T fields \-e frame.number \-e frame.time \-e eth.src \-e eth.dst \-e ip.src \-e ip.dst

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ tshark -r working/nitroba_working.pcap \
-Y 'ip.addr == 192.168.15.4' \
-T fields \
-e frame.number \
-e frame.time \
-e eth.src \
-e eth.dst \
-e ip.src \
-e ip.dst

18815   2008-07-22T00:29:51.677648000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    192.168.1.254
18816   2008-07-22T00:29:51.678746000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       192.168.1.254   192.168.15.4
18818   2008-07-22T00:29:53.706234000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    192.168.1.254
18819   2008-07-22T00:29:53.719600000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       192.168.1.254   192.168.15.4
18820   2008-07-22T00:29:53.728711000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    72.21.210.11
18821   2008-07-22T00:29:53.809869000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       72.21.210.11    192.168.15.4
18822   2008-07-22T00:29:53.812119000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    72.21.210.11
18823   2008-07-22T00:29:53.812853000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    72.21.210.11
18824   2008-07-22T00:29:53.902967000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       72.21.210.11    192.168.15.4
18826   2008-07-22T00:29:54.673425000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       72.21.210.11    192.168.15.4
18827   2008-07-22T00:29:54.675352000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       72.21.210.11    192.168.15.4
18828   2008-07-22T00:29:54.677607000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       72.21.210.11    192.168.15.4
18829   2008-07-22T00:29:54.678062000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    72.21.210.11
18830   2008-07-22T00:29:54.720260000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    72.21.210.11
18831   2008-07-22T00:29:54.761277000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       72.21.210.11    192.168.15.4
18832   2008-07-22T00:29:54.763342000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       72.21.210.11    192.168.15.4
18833   2008-07-22T00:29:54.765294000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    72.21.210.11
18834   2008-07-22T00:29:54.765499000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       72.21.210.11    192.168.15.4
18835   2008-07-22T00:29:54.793346000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    192.168.1.254
18836   2008-07-22T00:29:54.803603000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       72.21.210.11    192.168.15.4
18837   2008-07-22T00:29:54.805672000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       72.21.210.11    192.168.15.4
18838   2008-07-22T00:29:54.806130000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    72.21.210.11
18839   2008-07-22T00:29:54.814050000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       192.168.1.254   192.168.15.4
18840   2008-07-22T00:29:54.827712000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    69.22.167.225
18841   2008-07-22T00:29:54.828131000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    69.22.167.225
18842   2008-07-22T00:29:54.836991000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       69.22.167.225   192.168.15.4
18843   2008-07-22T00:29:54.838229000-0400      00:1d:d9:2e:4f:60       00:17:f2:e2:c0:ce       69.22.167.225   192.168.15.4
18844   2008-07-22T00:29:54.839565000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    69.22.167.225
18845   2008-07-22T00:29:54.839922000-0400      00:17:f2:e2:c0:ce       00:1d:d9:2e:4f:60       192.168.15.4    69.22.167.225
```

Search all cookies with Tshark

tshark -r working/nitroba_working.pcap \ -Y 'http.cookie' \ -T fields \ -e frame.number \ -e frame.time \ -e eth.src \ -e ip.src \ -e ip.dst \ -e http.host \ -e http.cookie

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ tshark -r working/nitroba_working.pcap \
-Y 'http.cookie' \
-T fields \
-e frame.number \
-e frame.time \
-e eth.src \
-e ip.src \
-e ip.dst \
-e http.host \
-e http.cookie
3       2008-07-21T21:51:07.114897000-0400       00:1d:d9:2e:4f:61       192.168.1.64    74.125.19.19    mail.google.com __utma=173272373.902839046.1185282466.1197590670.1198997636.12; __utmz=173272373.; GX=DQAAAG8AAAAfAfUyhQ_sggWzv9rJkh1qKAce_T4qWmRpGQYwFUhxRk5XWOcB5kG_dYKSKeKef2ubSJgML3DQn9xIH9gq6Uo-CjQ0RT_NTmK1FyrCAc9MneVvZjA_JUhHzUy6-zfWBK9yAv3Oh_qRLkKMVk4C1LA; S=gmail=1OFTCadh-U_8vClXyCDznQ:gmail_yj=ftV2ptv6CmMF75tsD4j6-Q:gmproxy=7ud63rWMnuM:gmproxy_yj=_PScfiZhkHs:gmproxy_yj_sub=Ur-7SoDI0mM; GMAIL_AT=xn3j32popno6mlbc3jnl1r7fac4285; gmailchat=mylady.ixchel@gmail.com/984626; rememberme=false; GMAIL_RTT=176; SID=DQAAAG4AAADmt3jT_B55uo7H5gJJvjtGEdgsnfF-jgm5y5PBFX6QttLC_3RMU4MHlLq6RJOPBnBUoHlfQxR-fy6G6DuGBvFyGIRjGxg8dBLpxSgd3Ps5Bk5cLdDEAwZz1tPW86OvBYD6FJikyjB0Kn6gLx9FHrBi; TZ=420; NID=12=F-75BlIcjxeMtos5dAcyIa31CUM_10jhW_MQSB4mWcB2XMjpiNqX56lqCBqLOtkoww7dK0k82w3rd95ce_cDES3Q28zG-mImP9VJXBT0wupzyGjKkWMzbqpvIMnjXO_e; PREF=ID=6d23dd9ef592e5cc:TM=1172981425:LM=1215674544:GM=1:S=37LPhMc_jLvRNQ3x
311     2008-07-21T21:51:24.341792000-0400       00:1d:d9:2e:4f:61       192.168.1.64    63.245.209.121  fxfeeds.mozilla.com     s_vi=[CS]v1|4875855000004A86-A000B2100007028[CE]; __utma=183859642.1072914972.1174698800.1215992549.1216616050.10; __utmz=183859642.1206154312.7.1.utmccn=(direct)|utmcsr=(direct)|utmcmd=(none); s_cc=true; s_sq=%5B%5BB%5D%5D; __utmc=183859642
363     2008-07-21T21:51:55.429615000-0400       00:1d:d9:2e:4f:61       192.168.1.64    69.17.116.124   www.phdcomics.com       __utmz=180107493.1211429768.1.1.utmccn=(direct)|utmcsr=(direct)|utmcmd=(none); __utma=180107493.400567268092162760 0.1211429766.1215843621.1216492615.12
425     2008-07-21T21:51:56.788372000-0400       00:1d:d9:2e:4f:61       192.168.1.64    69.17.116.124   www.phdcomics.com       __utmz=180107493.1211429768.1.1.utmccn=(direct)|utmcsr=(direct)|utmcmd=(none); t=1; __utmb=180107493; __utmc=180107493; __utma=180107493.400567268092162760 0.1211429766.1216492615.1216691540.13
451     2008-07-21T21:51:56.919151000-0400       00:1d:d9:2e:4f:61       192.168.1.64    69.17.116.124   www.phdcomics.com       __utmz=180107493.1211429768.1.1.utmccn=(direct)|utmcsr=(direct)|utmcmd=(none); t=1; __utmb=180107493; __utmc=180107493; __utma=180107493.400567268092162760 0.1211429766.1216492615.1216691540.13
530     2008-07-21T21:51:57.213378000-0400       00:1d:d9:2e:4f:61       192.168.1.64    69.17.116.124   www.phdcomics.com       __utmz=180107493.1211429768.1.1.utmccn=(direct)|utmcsr=(direct)|utmcmd=(none); t=1; __utmb=180107493; __utmc=180107493; __utma=180107493.400567268092162760 0.1211429766.1216492615.1216691540.13
636     2008-07-21T21:52:52.828818000-0400       00:1d:d9:2e:4f:61       192.168.1.64    17.250.236.65   phobos.apple.com        s_vi=[CS]v1|4374FC15000046AB-A000B5E00000091[CE]
646     2008-07-21T21:52:52.883703000-0400       00:1d:d9:2e:4f:61       192.168.1.64    69.22.167.214   ax.phobos.apple.com.edgesuite.net       s_vi=[CS]v1|4374FC15000046AB-A000B5E00000091[CE]
791     2008-07-21T21:52:53.420584000-0400       00:1d:d9:2e:4f:61       192.168.1.64    69.22.167.215   a1.phobos.apple.com     s_vi=[CS]v1|4374FC15000046AB-A000B5E00000091[CE]
792     2008-07-21T21:52:53.420720000-0400       00:1d:d9:2e:4f:61       192.168.1.64    69.22.167.215   a1.phobos.apple.com     s_vi=[CS]v1|4374FC15000046AB-A000B5E00000091[CE]
794     2008-07-21T21:52:53.421752000-0400       00:1d:d9:2e:4f:61       192.168.1.64    69.22.167.215   a1.phobos.apple.com     s_vi=[CS]v1|4374FC15000046AB-A000B5E00000091[CE]
818     2008-07-21T21:52:53.453564000-0400       00:1d:d9:2e:4f:61       192.168.1.64    69.22.167.215   a1.phobos.apple.com     s_vi=[CS]v1|4374FC15000046AB-A000B5E00000091[CE]
891     2008-07-21T21:52:53.546004000-0400       00:1d:d9:2e:4f:61       192.168.1.64    69.22.167.215   a1.phobos.apple.com     s_vi=[CS]v1|4374FC15000046AB-A000B5E00000091[CE]
1130    2008-07-21T21:53:04.787515000-0400       00:1d:d9:2e:4f:61       192.168.1.64    17.250.236.65   phobos.apple.com        s_vi=[CS]v1|4374FC15000046AB-A000B5E00000091[CE]; NSC_gjobodf-qpe11-mc=44c723352974
1138    2008-07-21T21:53:04.821862000-0400       00:1d:d9:2e:4f:61       192.168.1.64    69.22.167.214   ax.phobos.apple.com.edgesuite.net       s_vi=[CS]v1|4374FC15000046AB-A000B5E00000091[CE]; NSC_gjobodf-qpe11-mc=44c723352974
10011   2008-07-21T21:55:15.037980000-0400       00:1d:d9:2e:4f:61       192.168.1.64    69.28.176.53    image.weather.com       RMID=40070feb3e2cafc0; RMFD=011JxkvVO10P09|O10PgW|O10PiO|O10PiP; LocID=USCA0184; UserPreferences=3%7C+%7C0%7Creal%7Cfast%7C-1%7C-1%7C-1%7C-1%7C+%7C+%7C+%7C+%7C+%7C-1%7CBusinessTravel%7C+%7C+%7C+%7C%7C%7C%7C%7C%7C%7CUSCA0184%253A1%253ACarmel%2520Valley%252C%2520CA%255EUSNY0996%253A1%253ANew%2520York%252C%2520NY%255EUSVT0233%253A1%253AStowe%252C%2520VT%255E02138%253A4%253ACambridge%252C%2520MA%2520%252802138%2529%255EUSCA0987%253A1%253ASan%2520Francisco%252C%2520CA%255E93950%253A4%253APacific%2520Grove%252C%2520CA%2520%252893950%2529
10059   2008-07-21T21:56:08.365706000-0400       00:1d:d9:2e:4f:61       192.168.1.64    66.102.1.136    sb.google.com   rememberme=false; MPRF=H4sIAAAAAAAAAAvSEs0rb2OSNWA6_FtJjospNFiKyddRidM3MyUlJ7U4tUKL0zkxN6koMyU9teL3oxvfPyoBACEewC0zAAAA; __utmz=173272373.1212786908.7.7.utmccn=(referral)|utmcsr=video.google.com|utmcct=/|utmcmd=referral; __utmx=173272373.0000098319230992827 1:2:; __utma=173272373.1398790265.1198643490.1212695605.1212786908.7; PREF=ID=5bc5454c1feb167a:TM=1114643174:LM=1207629782:FV=2:GM=1:IG=3:S=D9lMGbYjUOnFAqMX; __utmx=173272373.0000098319230992827 1:1216134266:2592000; NID=12=S-gHu-_-QJPemWhmsR330PW_ozzEs1idu84akUyubU19CZGF9wnpg2ZzZCwLSxGk9758vkZEOiUZzMdQdjN8nRqyA00fA6jsD5jObVVAwfhO_HLRPhnvlTSsVyo47qe; TZ=420; GMAIL_RTT=61; S=static_files=Canv8wBryQ
10152   2008-07-21T21:56:22.006138000-0400       00:1d:d9:2e:4f:61       192.168.1.64    64.236.68.229   at.atwola.com   JEB2=488015656E651815C47C611E3F00016D8; badsrfi=V0aebd83e3731f127c488014597b22; atdemo=a3ZhZz1hZjI6dWEyODtrdnVnPTI7; atdses=0
10156   2008-07-21T21:56:22.028462000-0400       00:1d:d9:2e:4f:61       192.168.1.64    64.236.68.229   at.atwola.com   JEB2=488015656E651815C47C611E3F00016D8; badsrfi=V0aebd83e3731f127c488014597b22; atdemo=a3ZhZz1hZjI6dWEyODtrdnVnPTI7; atdses=0
10161   2008-07-21T21:56:22.056414000-0400       00:1d:d9:2e:4f:61       192.168.1.64    168.75.68.13    pix03.revsci.net        rsi_segs_1000000=pUPDRORORmfuIUoJyvOzCVgy/pjEkjhdzYx4wYfYjr0QZgJEHJs08tRf8WcUuLrQAFxcySqgq4lFtlR8qmZ5EYm2QQMyGpObby6s3VdDt3gztUlRA6gb/nT9MFQ==; NETID01=NFG4gDXL-goAAAAmaooA; NETSEGS_J05532=ea9107b1a5326a5b6J05532b0648a9ba5160b64883eed167bd7a96f7ebe3f50d8d22e1ed3dab0c9
10164   2008-07-21T21:56:22.090557000-0400       00:1d:d9:2e:4f:61       192.168.1.64    168.75.68.97    js.revsci.net   rsi_segs_1000000=pUPDRORORmfuIUoJyvOzCVgy/pjEkjhdzYx4wYfYjr0QZgJEHJs08tRf8WcUuLrQAFxcySqgq4lFtlR8qmZ5EYm2QQMyGpObby6s3VdDt3gztUlRA6gb/nT9MFQ==; NETID01=NFG4gDXL-goAAAAmaooA; NETSEGS_J05532=ea9107b1a5326a5b6J05532b0648a9ba5160b64883eed167bd7a96f7ebe3f50d8d22e1ed3dab0c9
10183   2008-07-21T21:56:22.191566000-0400       00:1d:d9:2e:4f:61       192.168.1.64    69.22.167.222   uac.advertising.com     ACID=bb780012154083550003!; BASE=d91EmxmALO3riarO5xdtHKKpS1/eppy1ErOEwPZs
```

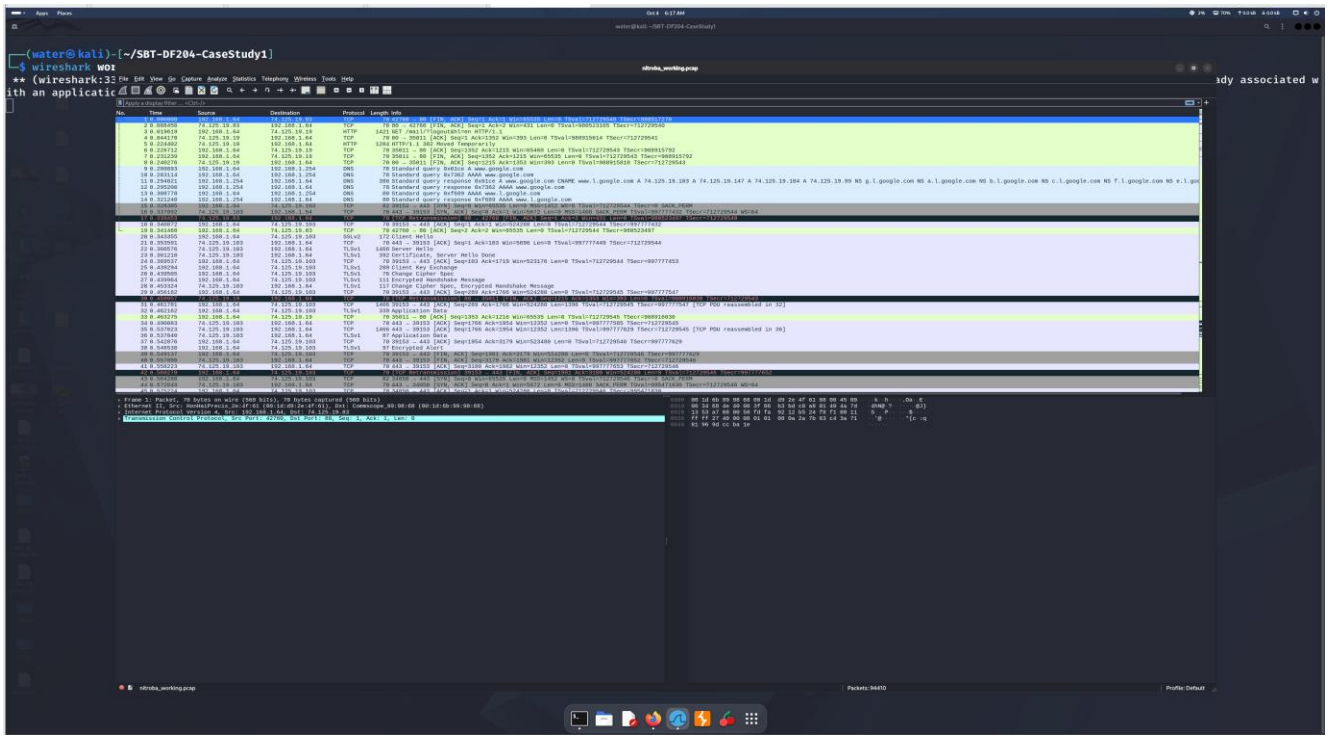
Finding TCP streams with Tshark

tshark -r working/nitroba_working.pcap \ -q -z conv,tcp

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ tshark -r working/nitroba_working.pcap \
-q -z conv,tcp
================================================================================
TCP Conversations
Filter:<No Filter>
                                                |       <-      | |       ->      | |     Total     |    Relative    |   Duration   |
                                                | Frames  Bytes | | Frames  Bytes | | Frames  Bytes |      Start     |              |
192.168.1.64:39710      <-> 208.111.148.6:80        3701 5,409 kB      2417 178 kB       6118 5,588 kB    118.152634000       30.7774
192.168.15.4:33148      <-> 74.125.15.159:80        3033 4,436 kB      1792 126 kB       4825 4,562 kB    10583.207131000      44.4780
192.168.15.4:34882      <-> 198.151.60.100:80        662 770 kB         532 109 kB       1194 880 kB      14189.336949000     513.5813
192.168.15.4:34884      <-> 198.151.60.100:80        574 662 kB         509 109 kB       1083 772 kB      14192.099396000     202.5940
192.168.15.4:32992      <-> 209.53.113.23:80         335 145 kB         668 147 kB       1003 293 kB      9938.591962000       13.7532
192.168.15.4:35956      <-> 69.22.167.247:80         585 851 kB         301 21 kB         886 873 kB      15130.506086000      67.0206
192.168.15.4:35742      <-> 204.2.133.48:80          431 589 kB         294 30 kB         725 620 kB      14921.650154000     442.2351
192.168.15.4:34742      <-> 69.22.167.239:80         407 427 kB         290 85 kB         697 513 kB      13871.077444000     606.6727
192.168.15.4:33944      <-> 69.22.167.215:80         419 609 kB         241 17 kB         660 626 kB      11054.696098000       0.9706
192.168.15.4:34118      <-> 69.22.167.214:80         397 576 kB         231 16 kB         628 592 kB      11144.435224000       0.9258
192.168.15.4:35264      <-> 209.123.229.231:80       399 571 kB         229 18 kB         628 590 kB      14566.048700000      16.8491
192.168.15.4:32916      <-> 199.93.58.125:80         326 433 kB         241 32 kB         567 465 kB      9848.318411000      537.7814
192.168.15.4:36558      <-> 209.17.73.28:80          366 501 kB         201 13 kB         567 514 kB      15545.470391000       1.2637
192.168.15.4:33988      <-> 69.22.167.214:80         343 497 kB         202 14 kB         545 512 kB      11069.614181000       0.7991
192.168.15.4:35138      <-> 66.211.160.61:80         309 270 kB         214 62 kB         523 332 kB      14449.948631000     1152.1452
192.168.15.4:33054      <-> 209.73.191.242:80        309 424 kB         200 22 kB         509 446 kB      10360.032501000      552.4429
192.168.15.4:35136      <-> 66.211.160.61:80         298 258 kB         204 59 kB         502 317 kB      14449.948172000     1152.1238
192.168.15.4:35804      <-> 74.125.19.17:80          274 235 kB         211 96 kB         485 331 kB      14986.306564000      242.2244
192.168.15.4:35446      <-> 216.113.185.92:80        243 164 kB         241 44 kB         484 209 kB      14775.800136000      827.4215
192.168.15.4:34744      <-> 69.22.167.239:80         236 299 kB         207 47 kB         443 347 kB      13871.708193000      162.9711
192.168.15.4:33710      <-> 69.22.167.215:80         260 375 kB         153 11 kB         413 386 kB      10997.264610000       0.6313
192.168.15.4:33652      <-> 69.22.167.215:80         261 375 kB         150 10 kB         411 386 kB      10981.447704000       0.6450
192.168.15.4:33220      <-> 69.26.160.17:80          210 263 kB         183 39 kB         393 302 kB      10761.402243000      34.9518
192.168.15.4:33602      <-> 69.22.167.214:80         244 350 kB         143 10 kB         387 361 kB      10972.614944000       0.5927
192.168.15.4:34656      <-> 65.182.192.74:80         222 302 kB         157 18 kB         379 321 kB      13824.918930000       6.1007
192.168.15.4:35544      <-> 66.166.239.194:80        199 243 kB         180 20 kB         379 264 kB      14802.548741000      23.7521
192.168.15.4:35702      <-> 216.113.185.92:80        190 124 kB         186 36 kB         376 160 kB      14869.272460000     678.2679
192.168.15.4:35746      <-> 204.2.133.48:80          219 287 kB         154 18 kB         373 306 kB      14921.726198000     442.1442
192.168.15.4:34888      <-> 198.151.61.175:80        193 188 kB         174 32 kB         367 221 kB      14194.757554000     493.1633
192.168.15.4:35266      <-> 209.123.229.231:80       219 308 kB         138 13 kB         357 322 kB      14566.049171000      16.8497
192.168.15.4:35788      <-> 66.211.160.22:80         205 175 kB         141 41 kB         346 217 kB      14960.424019000     642.2345
192.168.15.4:34072      <-> 69.22.167.214:80         208 300 kB         123 9,060 bytes    331 309 kB      11097.790719000       0.5942
192.168.15.4:34730      <-> 66.39.211.225:80         177 218 kB         151 18 kB         328 237 kB      13850.775442000       2.6272
192.168.15.4:35074      <-> 160.143.188.82:80        189 256 kB         132 12 kB         321 269 kB      16398.020424000      36.5916
192.168.15.4:35704      <-> 216.113.185.92:80        158 105 kB         152 29 kB         310 135 kB      14869.283423000     678.2063
192.168.15.4:35144      <-> 216.113.185.92:80        158 100 kB         150 26 kB         308 127 kB      14449.993112000     294.9943
192.168.15.4:35146      <-> 216.113.185.92:80        150 93 kB          147 25 kB         297 119 kB      14449.993757000     352.9590
192.168.15.4:34712      <-> 65.182.192.74:80         167 224 kB         126 25 kB         293 249 kB      13832.232381000      19.9172
192.168.15.4:34894      <-> 198.151.61.175:80        150 140 kB         137 26 kB         287 167 kB      14195.827060000     492.0956
192.168.15.4:33608      <-> 69.22.167.214:80         178 255 kB         106 7,867 bytes    284 262 kB      10973.970616000       0.4629
192.168.15.4:35818      <-> 204.2.133.41:80          153 204 kB         101 9,384 bytes    254 213 kB      14987.890094000       1.2699
192.168.15.4:34274      <-> 72.14.253.125:5222       122 44 kB          127 27 kB         249 71 kB       11456.138444000     910.0153
192.168.1.64:35700      <-> 65.212.121.21:80         133 187 kB         108 7,102 bytes    241 195 kB      245.220290000        3.9386
192.168.1.64:45685      <-> 64.233.187.190:80        158 215 kB          82 6,776 bytes    240 222 kB      6879.192469000     155.3080
192.168.15.4:34548      <-> 63.111.24.75:80          141 184 kB          98 6,462 bytes    239 190 kB      12845.546584000       2.2699
192.168.15.4:32822      <-> 8.12.217.125:80          127 145 kB         108 14 kB         235 160 kB      9528.286259000      542.9038
```

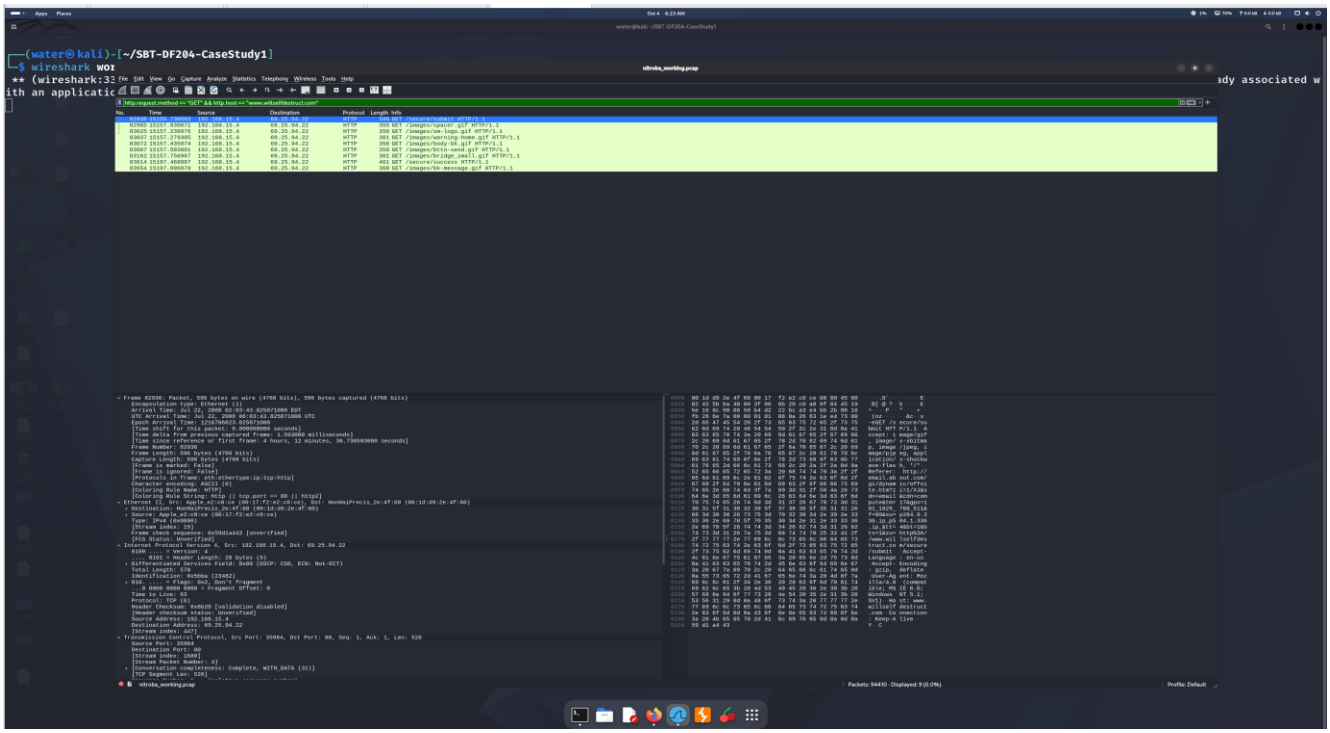
Open the PCAP in Wireshark

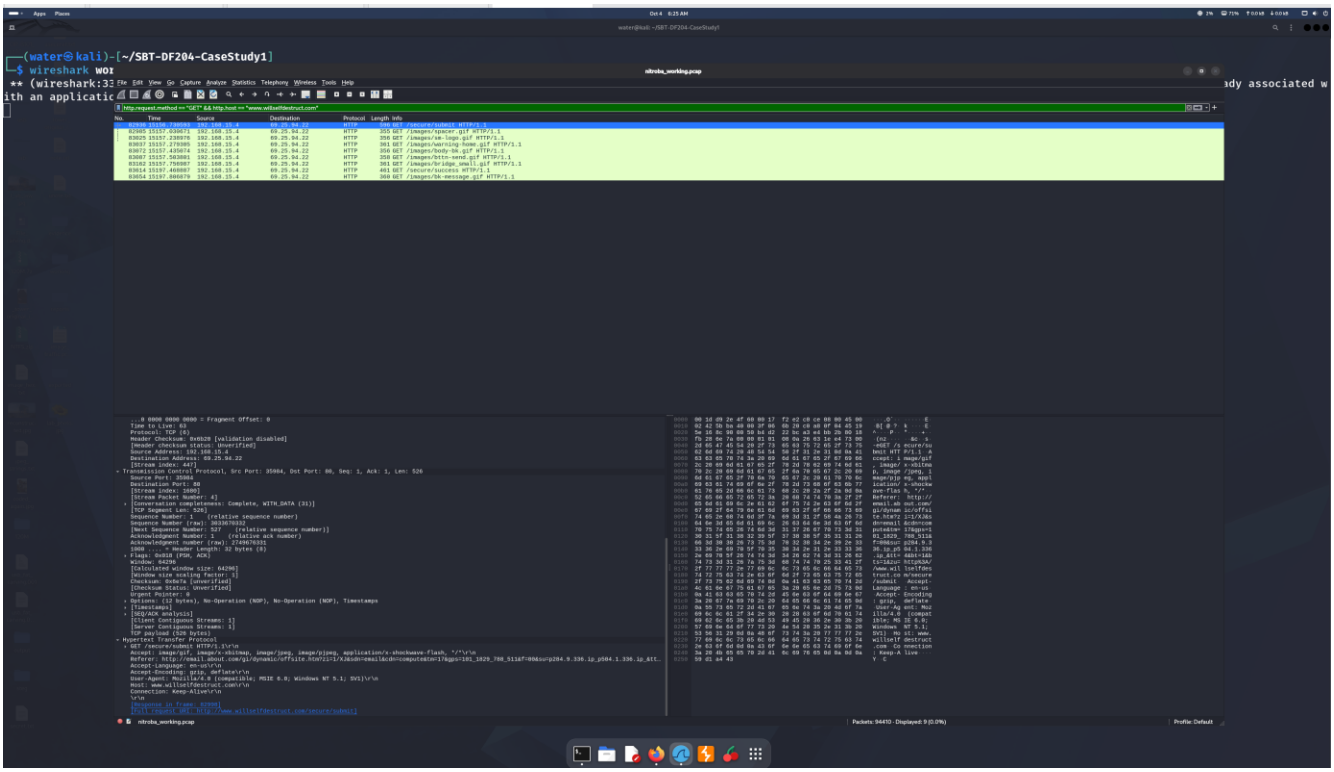
wireshark working/nitroba_working.pcap



Finding the willselfdestruct.com traffic on the wire shark

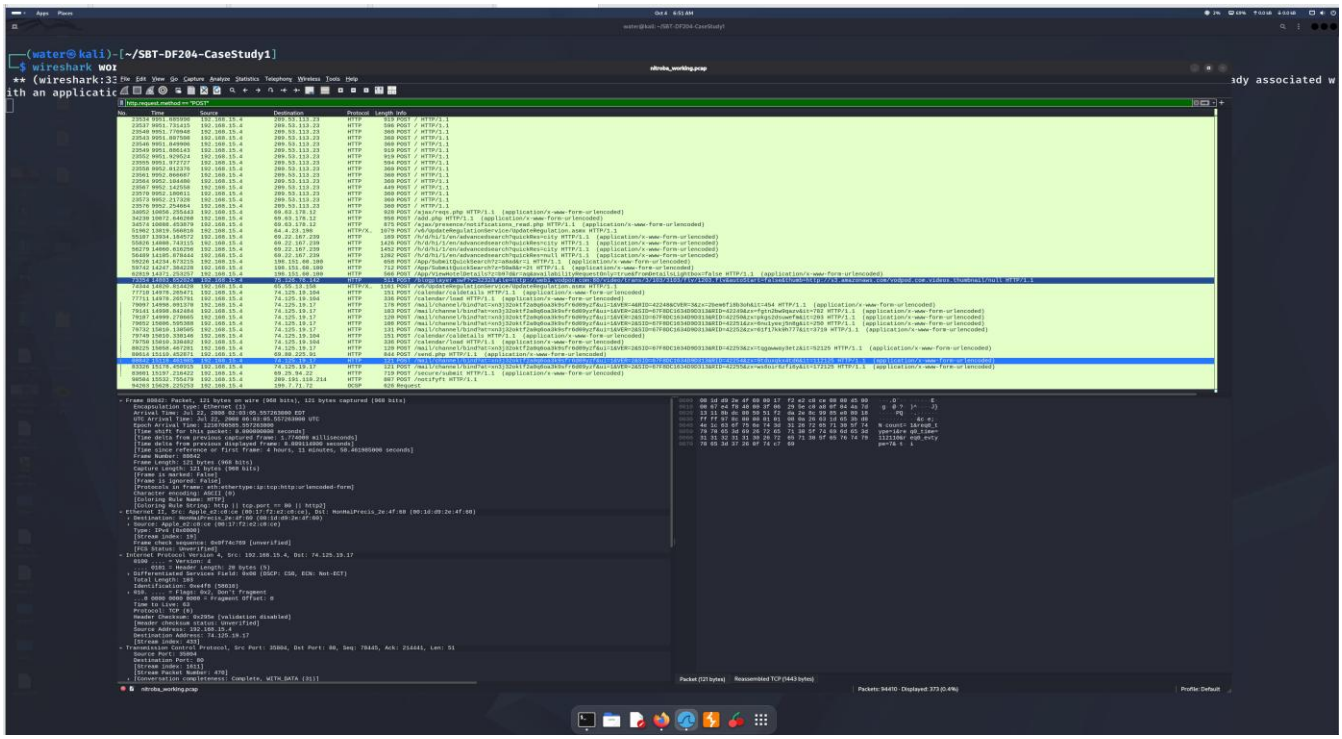
http.request.method == "GET" && http.host == "www.willselfdestruct.com"

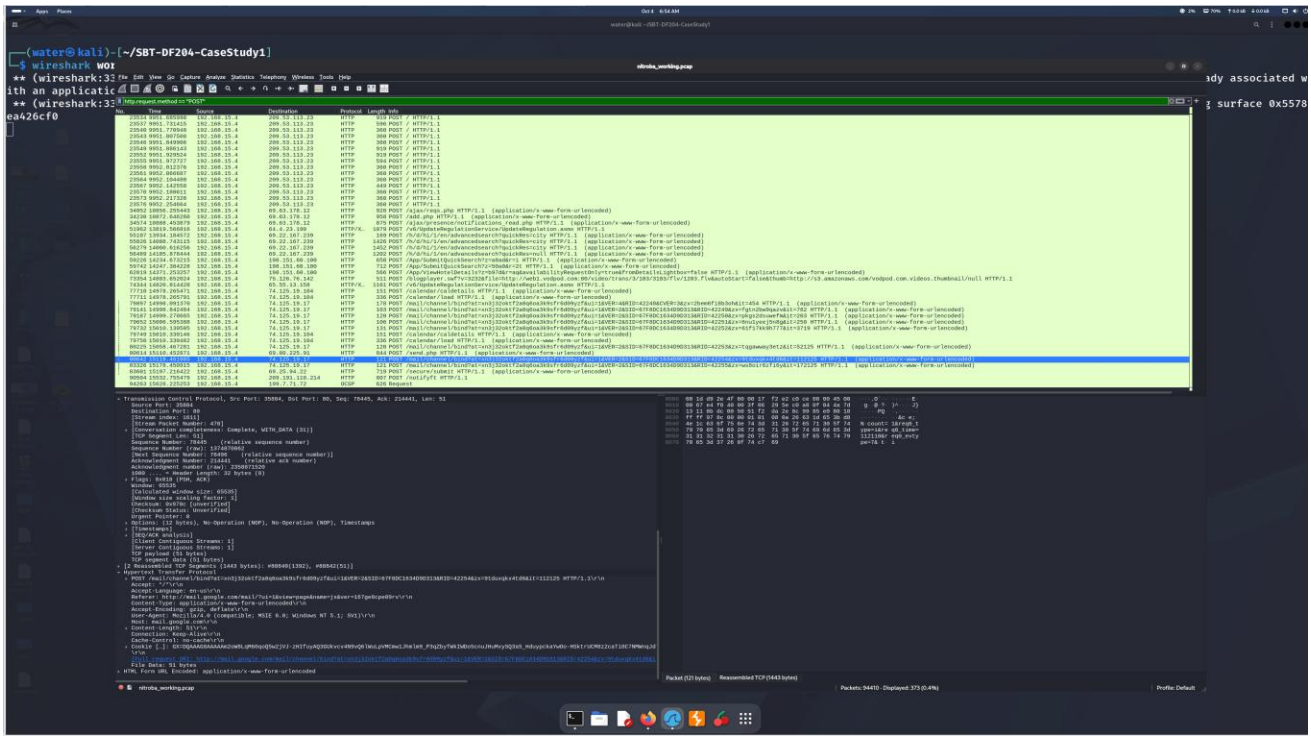




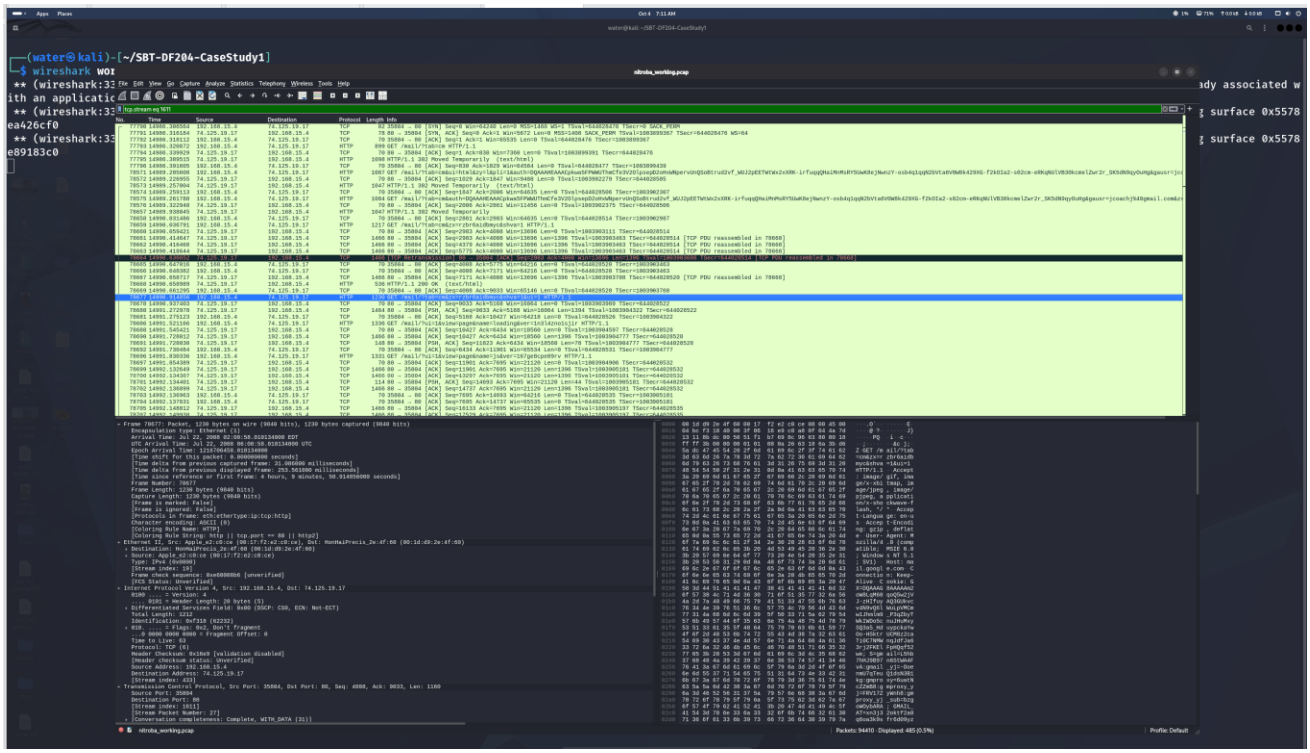
Finding the HTTP POST

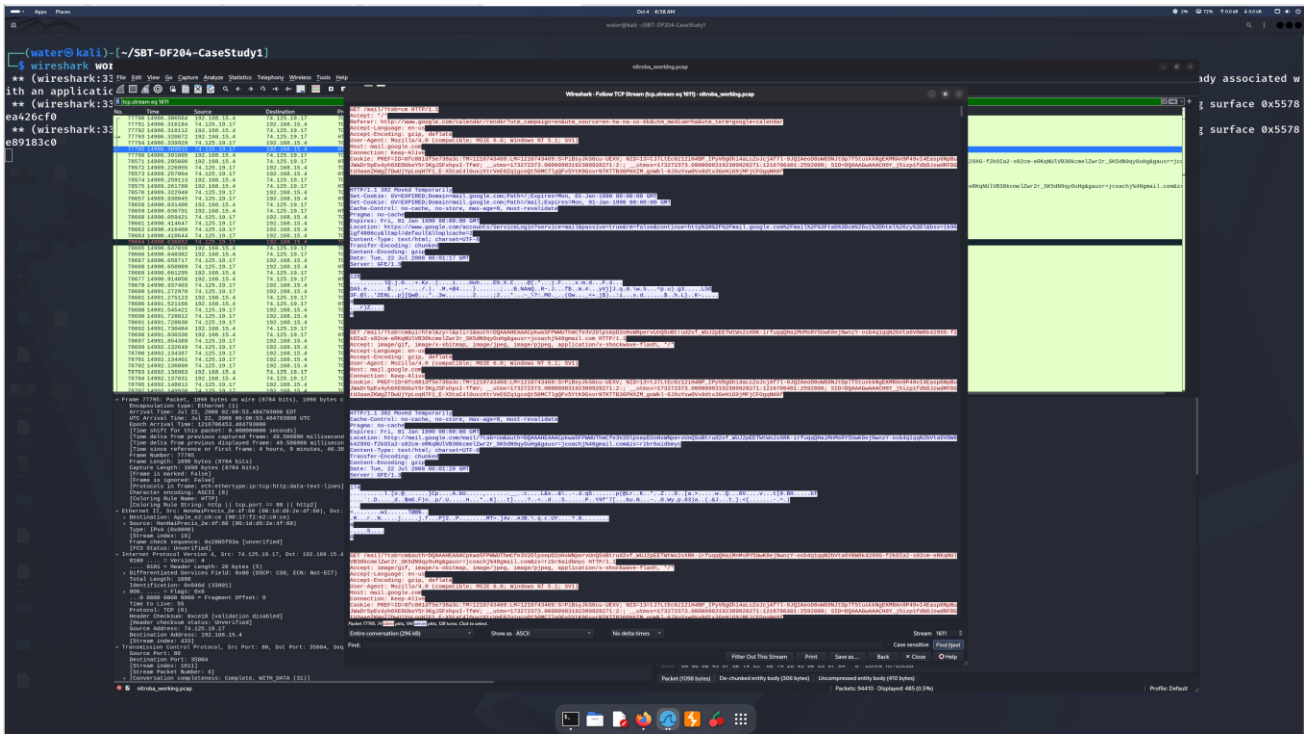
http.request.method == "POST"



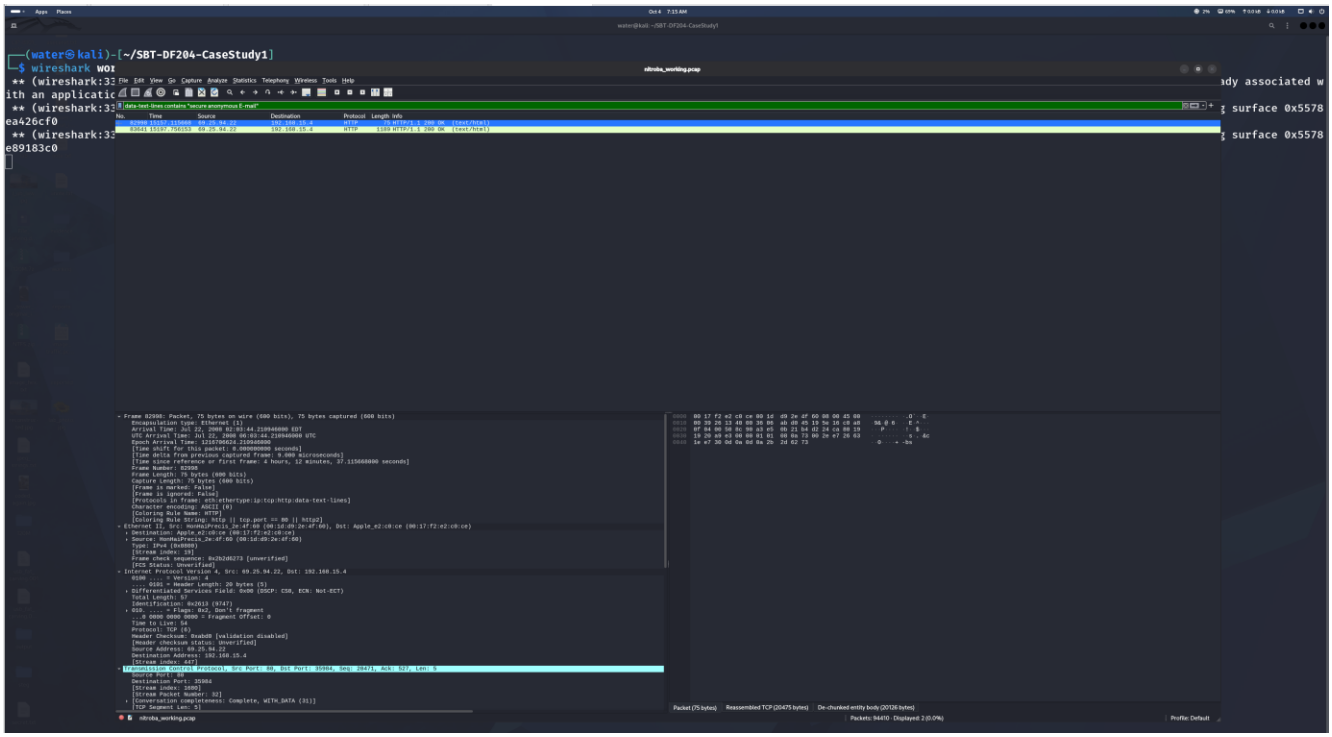


For tcp.stream eq 1611





Searching for the message content (data-text-lines contains "secure anonymous E-mail")

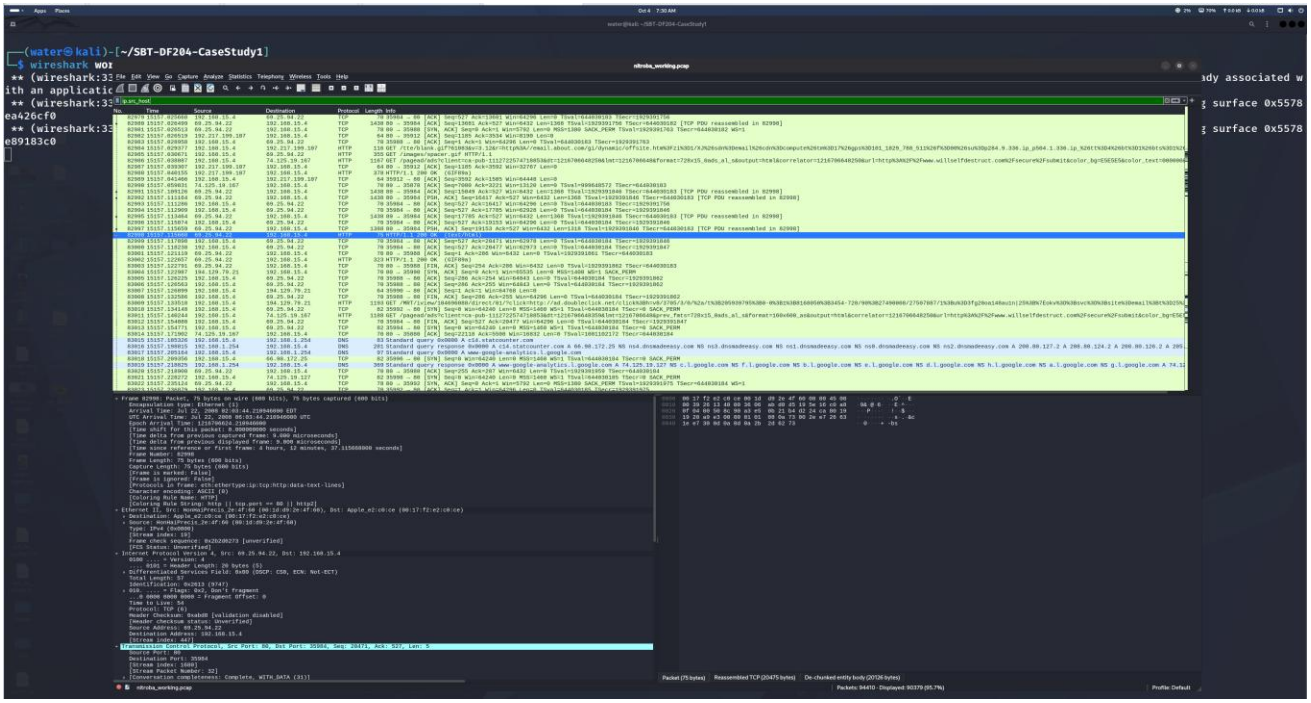


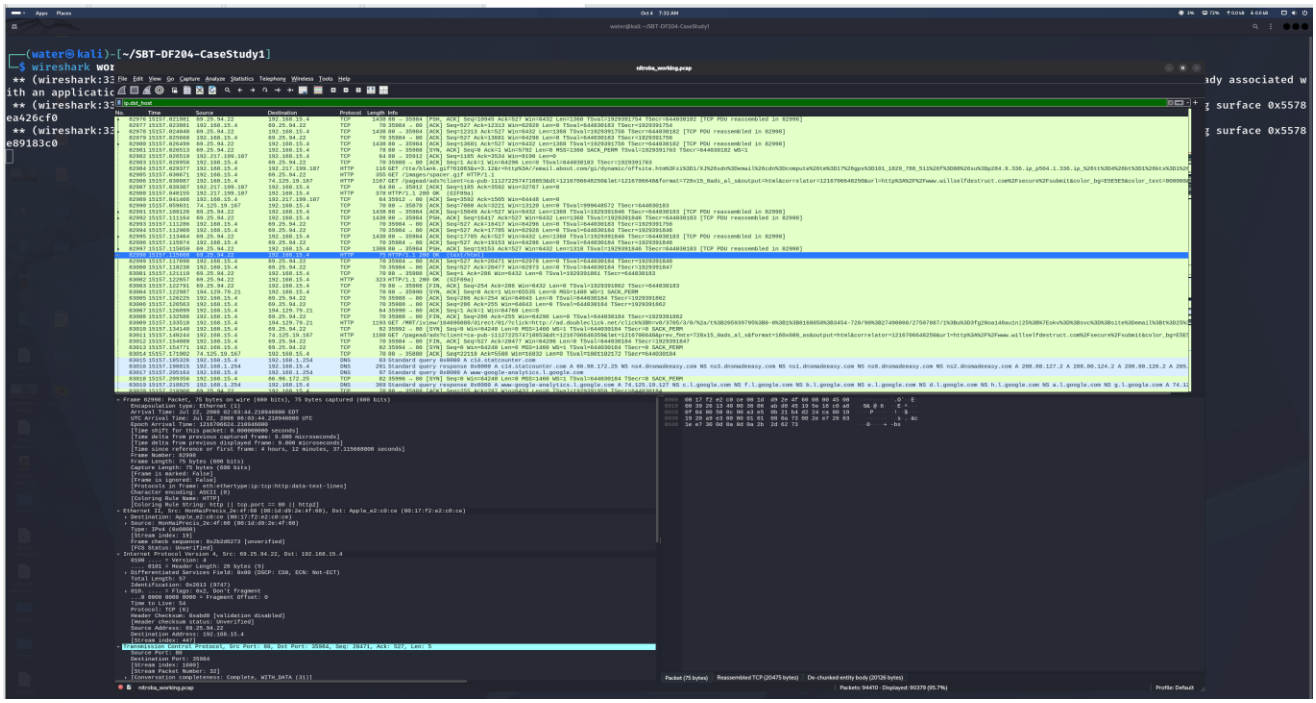
Identifying the client and service IP (ip.src == CLIENT_IP && ip.dst == SERVICE_IP)

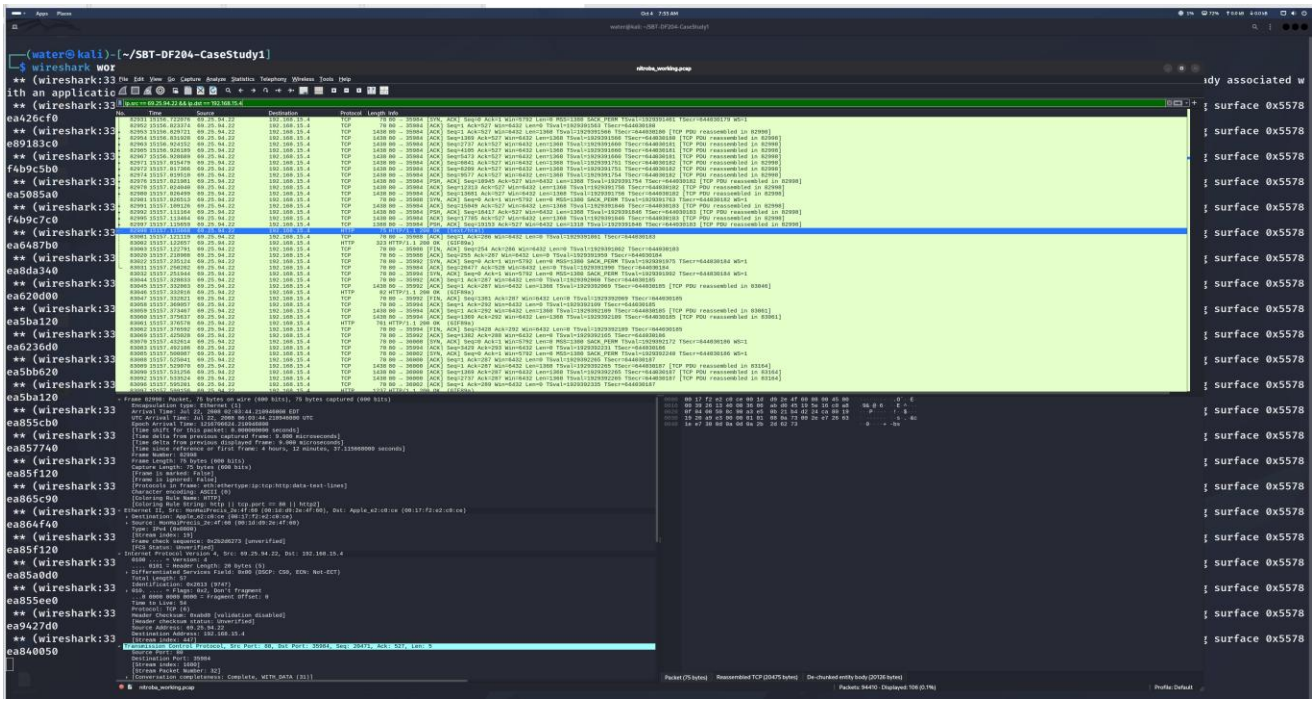
Source IP: 192.168.15.4

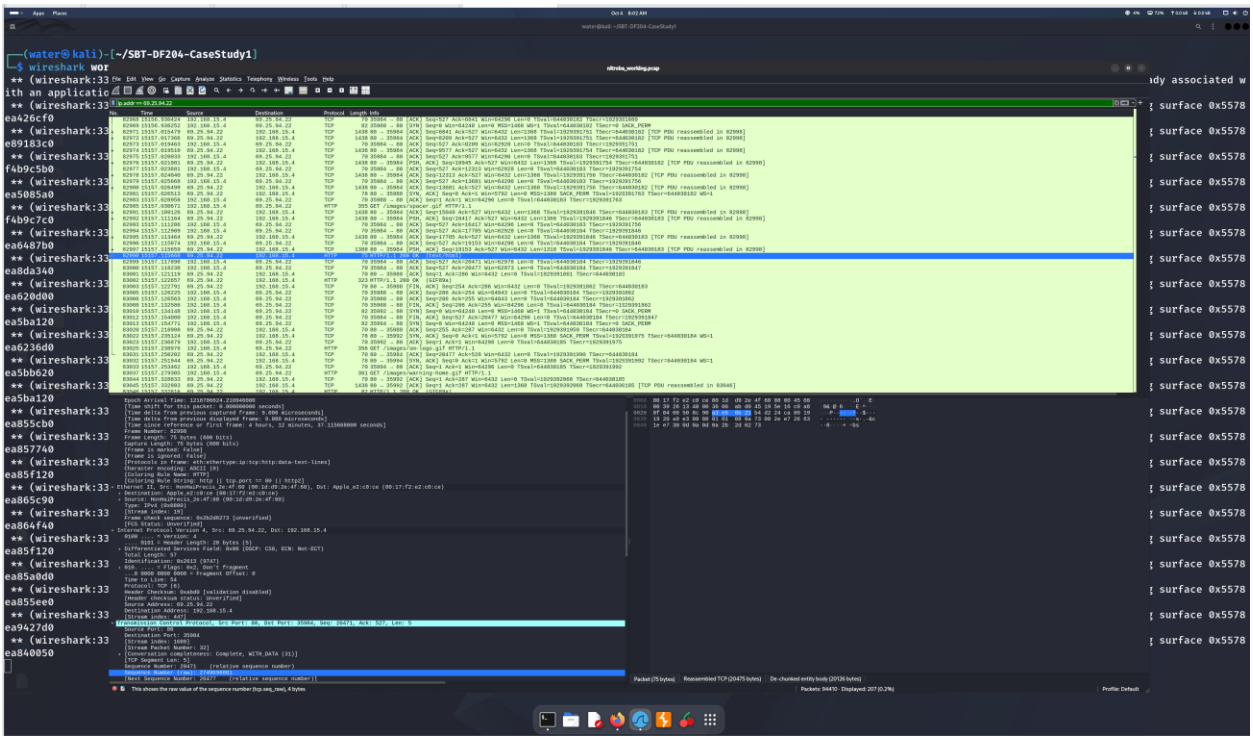
Destination IP address: 69.25.94.22

Protocol:HTTP



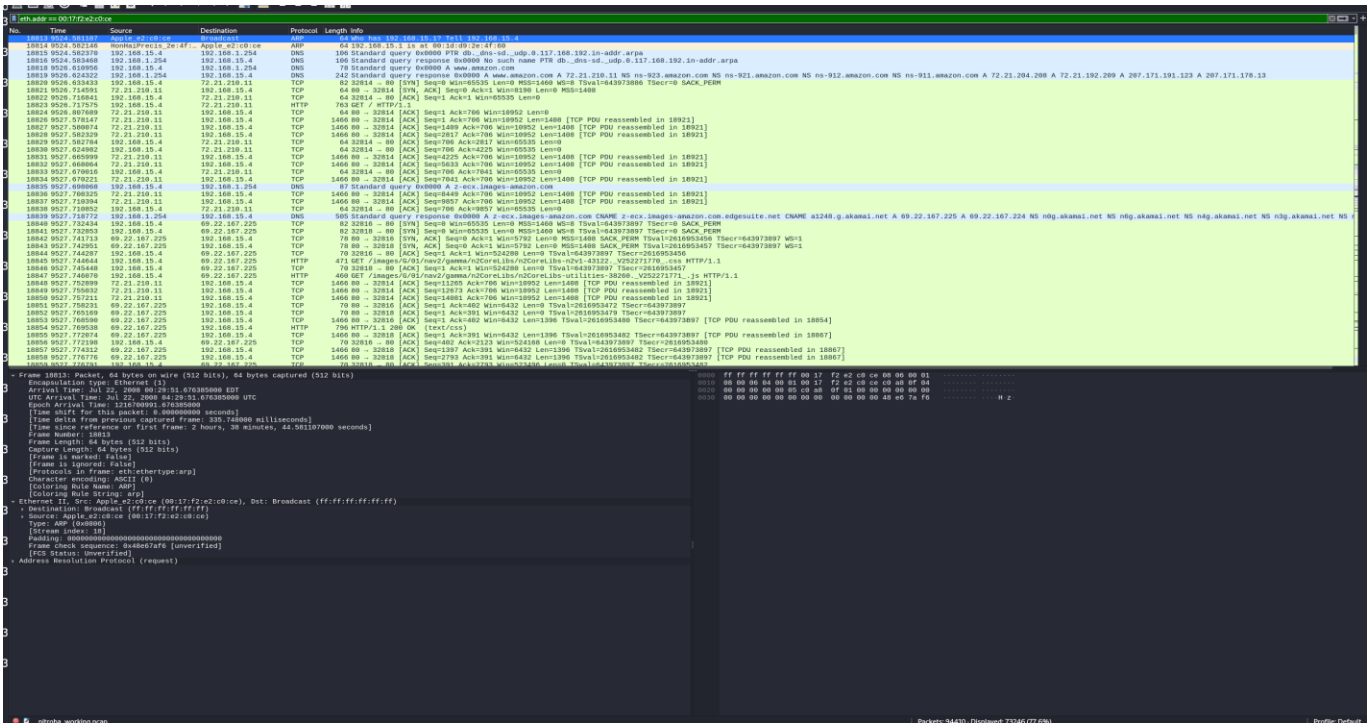




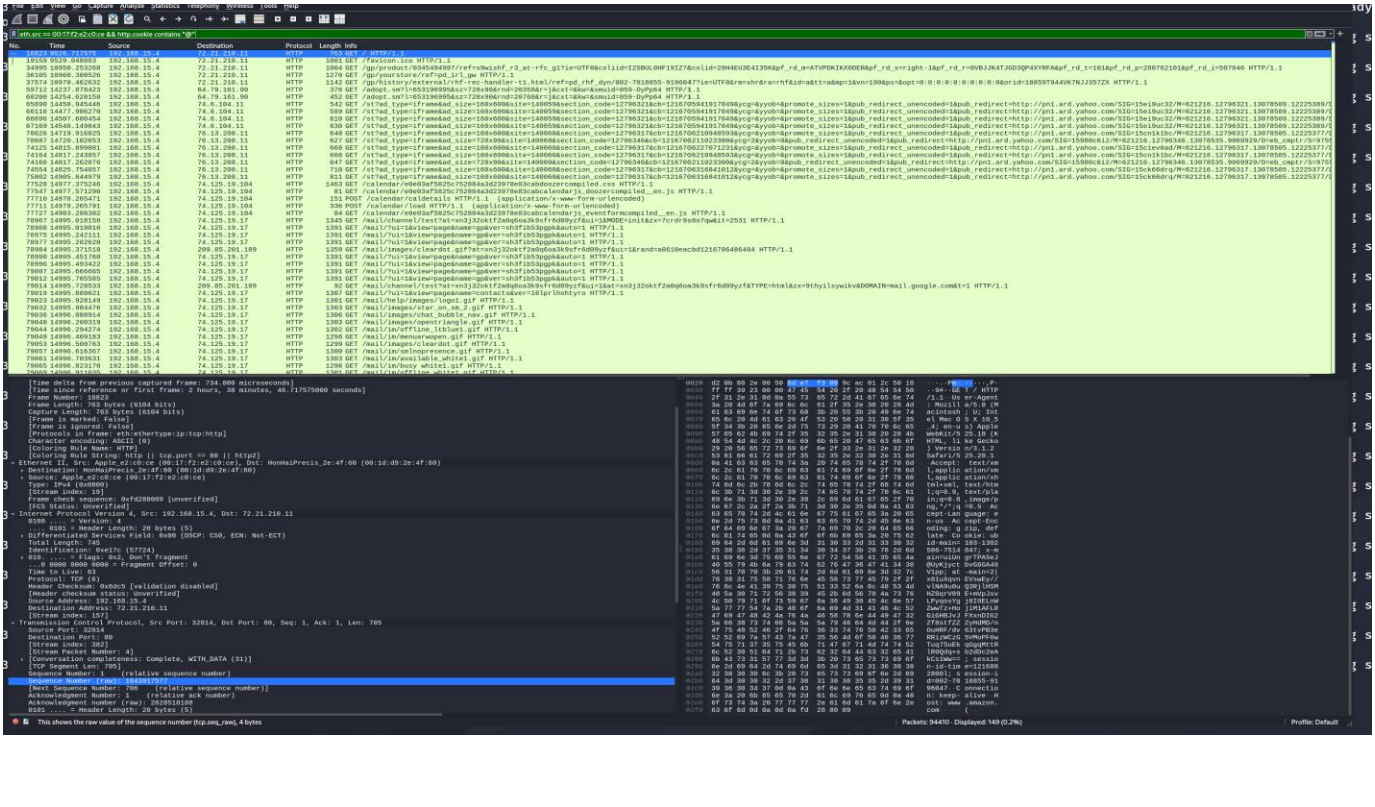


eth.addr == <client_mac> (eth.addr == 00:17:f2:e2:c0:ce)

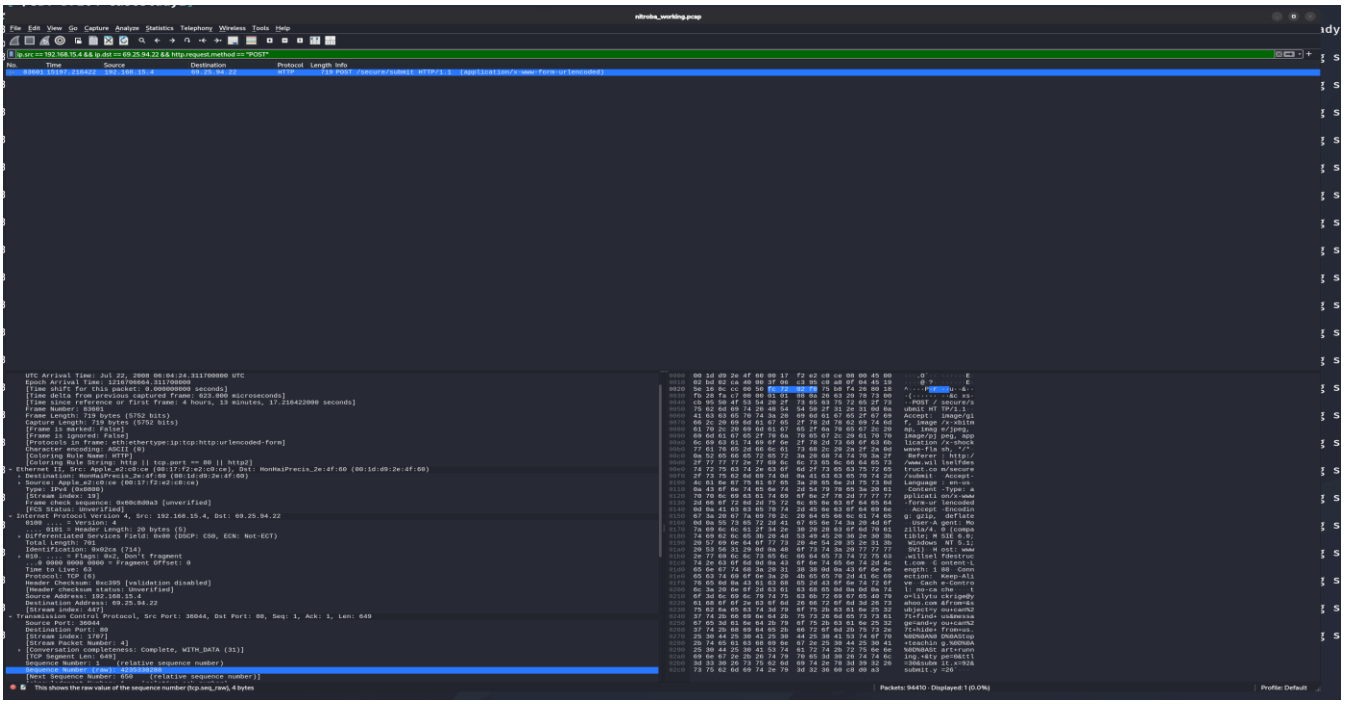
The source ip address is192.168.15.4 and mac address 00:17:f2:e2:c0:ce, an apple device.



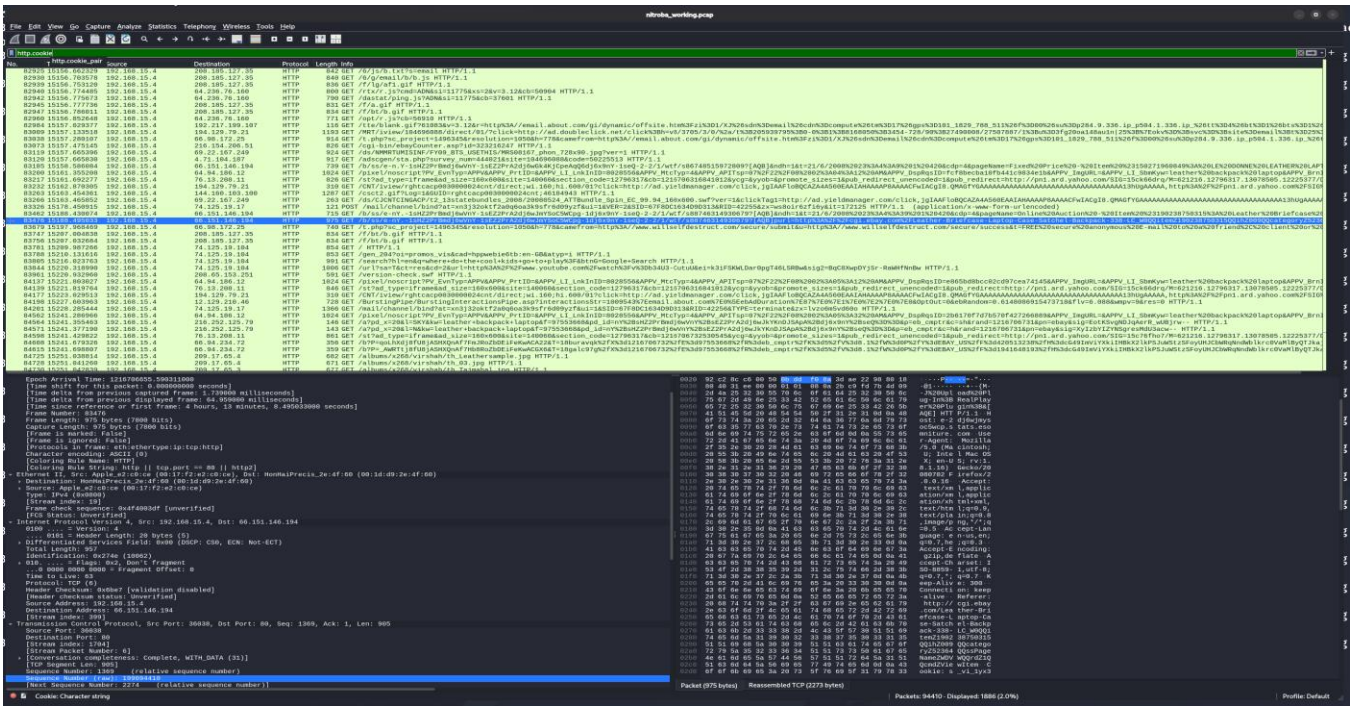
eth.src == <client_mac> && http.cookie contains "@" (eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@")

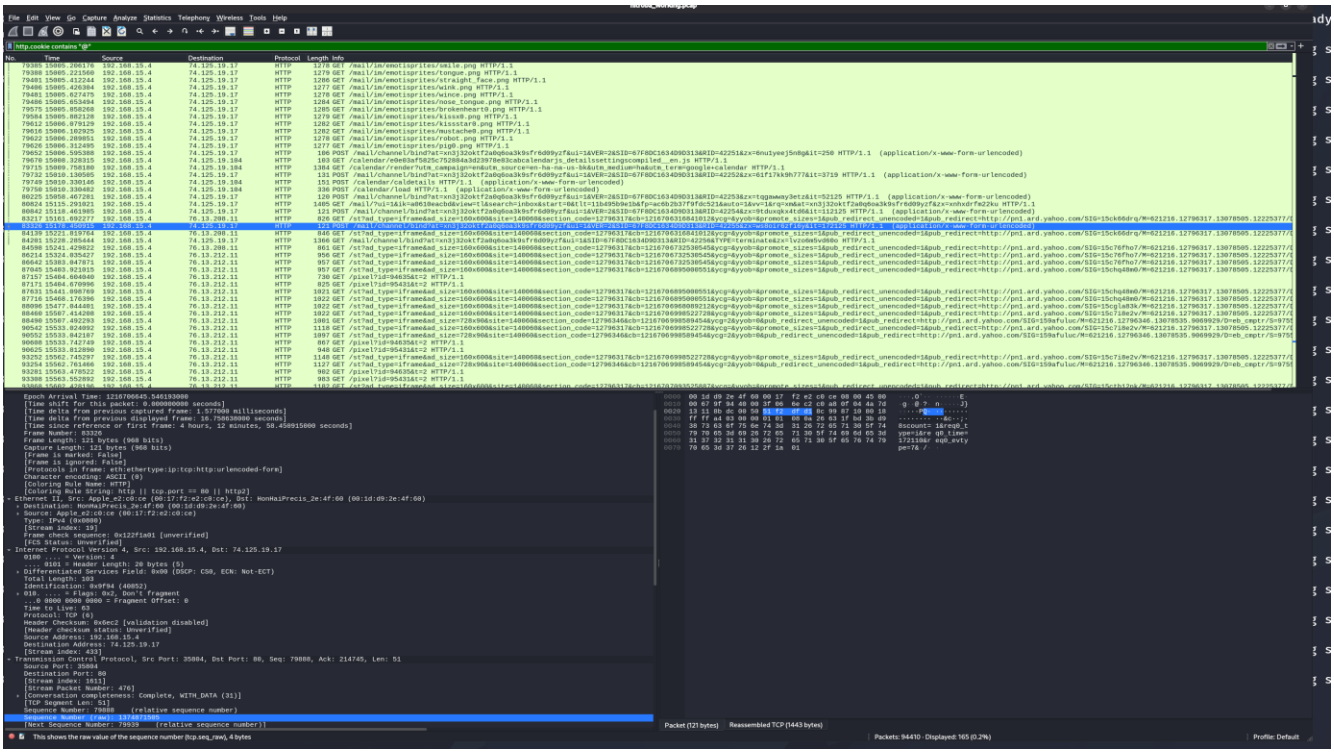


ip.src == <client_ip> && ip.dst == <service_ip> && http.request.method == "POST" (ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 && http.request.method == "POST")



Checking all cookies (http.cookie) (http.cookie contains "@") (http.set_cookie)




File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http.cookie contains "@"
No. Time Source Destination Protocol Length Info
79385 15905.266176 192.168.15.4 74.125.19.17 HTTP 1278 GET /mail/im/emotisprites/smile.png HTTP/1.1
79388 15905.221560 192.168.15.4 74.125.19.17 HTTP 1279 GET /mail/im/emotisprites/tongue_png HTTP/1.1
79481 15905.412244 192.168.15.4 74.125.19.17 HTTP 1286 GET /mail/im/emotisprites/straight_face.png HTTP/1.1
79486 15905.426304 192.168.15.4 74.125.19.17 HTTP 1277 GET /mail/im/emotisprites/wink.png HTTP/1.1
79481 15905.627475 192.168.15.4 74.125.19.17 HTTP 1278 GET /mail/im/emotisprites/wince.png HTTP/1.1
79486 15905.653494 192.168.15.4 74.125.19.17 HTTP 1284 GET /mail/im/emotisprites/nose_tongue.png HTTP/1.1
79575 15905.858268 192.168.15.4 74.125.19.17 HTTP 1285 GET /mail/im/emotisprites/brokenheart0.png HTTP/1.1
79584 15905.862128 192.168.15.4 74.125.19.17 HTTP 1279 GET /mail/im/emotisprites/kiss0.png HTTP/1.1
79612 15906.079129 192.168.15.4 74.125.19.17 HTTP 1282 GET /mail/im/emotisprites/kisstar0.png HTTP/1.1
79615 15906.102925 192.168.15.4 74.125.19.17 HTTP 1282 GET /mail/im/emotisprites/mustached.png HTTP/1.1
79622 15906.289051 192.168.15.4 74.125.19.17 HTTP 1278 GET /mail/im/emotisprites/robot.png HTTP/1.1
79626 15906.312495 192.168.15.4 74.125.19.17 HTTP 1277 GET /mail/im/emotisprites/pig0.png HTTP/1.1
79652 15908.595388 192.168.15.4 74.125.19.17 HTTP 106 POST /mail/channel/bind?at=xn3j32okTf2aDq6oa3k9sfr6d09yzf&ui=1&VER=2&SID=67F0DC1634D9D313&RID=42251&zx=6nu1yeeJSn8g&it=250 HTTP/1.1 (application/x-www-form-urlencoded)
79670 15908.328315 192.168.15.4 74.125.19.104 HTTP 103 GET /calendar/ebe0baf5825c752884a3d23978e83cabcalendarjs_detailsettingscompiled__en.js HTTP/1.1
79715 15909.758180 192.168.15.4 74.125.19.104 HTTP 1384 GET /calendar/render?utm_campaign=en&utm_source=en-ha-na-us-bk&utm_medium=ha&utm_term=google+calendar HTTP/1.1
79732 15910.130505 192.168.15.4 74.125.19.17 HTTP 131 POST /mail/channel/bind?at=xn3j32okTf2aDq6oa3k9sfr6d09yzf&ui=1&VER=2&SID=67F0DC1634D9D313&RID=42252&zx=6ifi7kk9h777&it=3719 HTTP/1.1 (application/x-www-form-urlencoded)
79749 15910.330146 192.168.15.4 74.125.19.104 HTTP 151 POST /calendar/caldetails HTTP/1.1 (application/x-www-form-urlencoded)
79756 15910.330402 192.168.15.4 74.125.19.104 HTTP 336 POST /calendar/load HTTP/1.1 (application/x-www-form-urlencoded)
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80824 15115.291021 192.168.15.4 74.125.19.17 HTTP 1405 GET 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vqvf=1&rvqvg=1&rvqvh=1&rvqvi=1&rvqvj=1&rvqvk=1&rvqvl=1&rvqvm=1&rvqvn=1&rvqvo=1&rvqvp=1&rvqvq=1&rvqvr=1&rvqvs=1&rvqvt=1&rvqvu=1&rvqvv=1&rvqvw=1&rvqvx=1&rvqvy=1&rvqvz=1&rvqwa=1&rvqwb=1&rvqwc=1&rvqwd=1&rvqwe=1&rvqwf=1&rvqwg=1&rvqwh=1&rvqwi=1&rvqwj=1&rvqwk=1&rvqwl=1&rvqwm=1&rvqwn=1&rvqwo=1&rvqwp=1&rvqwq=1&rvqwr=1&rvqws=1&rvqwt=1&rvqwu=1&rvqwv=1&rvqww=1&rvqwx=1&rvqwy=1&rvqwz=1&rvqxa=1&rvqxb=1&rvqxc=1&rvqxd=1&rvqxe=1&rvqxf=1&rvqxg=1&rvqxh=1&rvqxi=1&rvqxj=1&rvqxk=1&rvqxl=1&rvqxm=1&rvqxn=1&rvqxo=1&rvqxp=1&rvqxq=1&rvqxr=1&rvqxs=1&rvqxt=1&rvqxu=1&rvqxv=1&rvqxw=1&rvqxx=1&rvqxy=1&rvqxz=1&rvqya=1&rvqyb=1&rvqyc=1&rvqyd=1&rvqye=1&rvqyf=1&rvqyg=1&rvqyh=1&rvqyi=1&rvqyj=1&rvqyk=1&rvqyl=1&rvqym=1&rvqyn=1&rvqyo=1&rvqyp=1&rvqyq=1&rvqyr=1&rvqys=1&rvqyt=1&rvqyu=1&rvqyv=1&rvqyw=1&rvqyx=1&rvqyy=1&rvqyz=1&rvqza=1&rvqzb=1&rvqzc=1&rvqzd=1&rvqze=1&rvqzf=1&rvqzg=1&rvqzh=1&rvqzi=1&rvqzj=1&rvqzk=1&rvqzl=1&rvqzm=1&rvqzn=1&rvqzo=1&rvqzp=1&rvqzq=1&rvqzr=1&rvqzs=1&rvqzt=1&rvqzu=1&rvqzv=1&rvqzw=1&rvqzx=1&rvqzy=1&rvqzz=1&rvraa=1&rvrab=1&rvrac=1&rvrad=1&rvrae=1&rvraf=1&rvrag=1&rvrah=1&rvrai=1&rvraj=1&rvrak=1&rvral=1&rvram=1&rvran=1&rvrao=1&rvrap=1&rvraq=1&rvrar=1&rvras=1&rvrat=1&rvrau=1&rvrav=1&rvraw=1&rvrax=1&rvray=1&rvraz=1&rvrba=1&rvrbb=1&rvrbc=1&rvrbd=1&rvrbe=1&rvrbf=1&rvrbg=1&rvrbh=1&rvrbi=1&rvrbj=1&rvrbk=1&rvrbl=1&rvrbm=1&rvrbn=1&rvrbo=1&rvrbp=1&rvrbq=1&rvrbr=1&rvrbs=1&rvrbt=1&rvrbu=1&rvrbv=1&rvrbw=1&rvrbx=1&rvrby=1&rvrbz=1&rvrca=1&rvrcb=1&rvrcc=1&rvrcd=1&rvrce=1&rvrcf=1&rvrcg=1&rvrch=1&rvrci=1&rvrcj=1&rvrck=1&rvrcl=1&rvrcm=1&rvrcn=1&rvrco=1&rvrcp=1&rvrcq=1&rvrcr=1&rvrcs=1&rvrct=1&rvrcu=1&rvrcv=1&rvrcw=1&rvrcx=1&rvrcy=1&rvrcz=1&rvrda=1&rvrdb=1&rvrdc=1&rvrdd=1&rvrde=1&rvrdf=1&rvrdg=1&rvrdh=1&rvrdi=1&rvrdj=1&rvrdk=1&rvrdl=1&rvrdm=1&rvrdn=1&rvrdo=1&rvrdp=1&rvrdq=1&rvrdr=1&rvrds=1&rvrdt=1&rvrdu=1&rvrdv=1&rvrdw=1&rvrdx=1&rvrdy=1&rvrdz=1&rvrea=1&rvreb=1&rvrec=1&rvred=1&rvree=1&rvref=1&rvreg=1&rvreh=1&rvrei=1&rvrej=1&rvrek=1&rvrel=1&rvrem=1&rvren=1&rvreo=1&rvrep=1&rvreq=1&rvrer=1&rvres=1&rvret=1&rvreu=1&rvrev=1&rvrew=1&rvrex=1&rvrey=1&rvrez=1&rvrfa=1&rvrfb=1&rvrfc=1&rvrfd=1&rvrfe=1&rvrff=1&rvrfg=1&rvrfh=1&rvrfi=1&rvrfj=1&rvrfk=1&rvrfl=1&rvrfm=1&rvrfn=1&rvrfo=1&rvrfp=1&rvrfq=1&rvrfr=1&rvrfs=1&rvrft=1&rvrfu=1&rvrfv=1&rvrfw=1&rvrfx=1&rvrfy=1&rvrfz=1&rvrga=1&rvrgb=1&rvrgc=1&rvrgd=1&rvrge=1&rvrgf=1&rvrgg=1&rvrgh=1&rvrgi=1&rvrgj=1&rvrgk=1&rvrgl=1&rvrgm=1&rvrgn=1&rvrgo=1&rvrgp=1&rvrgq=1&rvrgr=1&rvrgs=1&rvrgt=1&rvrgu=1&rvrgv=1&rvrgw=1&rvrgx=1&rvrgy=1&rvrgz=1&rvrha=1&rvrhb=1&rvrhc=1&rvrhd=1&rvrhe=1&rvrhf=1&rvrhg=1&rvrhh=1&rvrhi=1&rvrhj=1&rvrhk=1&rvrhl=1&rvrhm=1&rvrhn=1&rvrho=1&rvrhp=1&rvrhq=1&rvrhr=1&rvrhs=1&rvrht=1&rvrhu=1&rvrhv=1&rvrhw=1&rvrhx=1&rvrhy=1&rvrhz=1&rvria=1&rvrib=1&rvric=1&rvrid=1&rvrie=1&rvrif=1&rvrig=1&rvrih=1&rvrii=1&rvrij=1&rvrik=1&rvril=1&rvrim=1&rvrin=1&rvrio=1&rvrip=1&rvriq=1&rvrir=1&rvris=1&rvrit=1&rvriu=1&rvriv=1&rvriw=1&rvrix=1&rvriy=1&rvriz=1&rvrja=1&rvrjb=1&rvrjc=1&rvrjd=1&rvrje=1&rvrjf=1&rvrjg=1&rvrjh=1&rvrji=1&rvrjj=1&rvrjk=1&rvrjl=1&rvrjm=1&rvrjn=1&rvrjo=1&rvrjp=1&rvrjq=1&rvrjr=1&rvrjs=1&rvrjt=1&rvrju=1&rvrjv=1&rvrjw=1&rvrjx=1&rvrjy=1&rvrjz=1&rvrka=1&rvrkb=1&rvrkc=1&rvrkd=1&rvrke=1&rvrkf=1&rvrkg=1&rvrkh=1&rvrki=1&rvrkj=1&rvrkk=1&rvrkl=1&rvrkm=1&rvrkn=1&rvrko=1&rvrkp=1&rvrkq=1&rvrkr=1&rvrks=1&rvrkt=1&rvrku=1&rvrkv=1&rvrkw=1&rvrkx=1&rvrky=1&rvrkz=1&rvrla=1&rvrlb=1&rvrlc=1&rvrld=1&rvrle=1&rvrlf=1&rvrlg=1&rvrlh=1&rvrli=1&rvrlj=1&rvrlk=1&rvrll=1&rvrlm=1&rvrln=1&rvrlo=1&rvrlp=1&rvrlq=1&rvrlr=1&rvrls=1&rvrlt=1&rvrlu=1&rvrlv=1&rvrlw=1&rvrlx=1&rvrly=1&rvrlz=1&rvrma=1&rvrmb=1&rvrmc=1&rvrmd=1&rvrme=1&rvrmf=1&rvrmg=1&rvrmh=1&rvrmi=1&rvrmj=1&rvrmk=1&rvrml=1&rvrmm=1&rvrmn=1&rvrmo=1&rvrmp=1&rvrmq=1&rvrmr=1&rvrms=1&rvrmt=1&rvrmu=1&rvrmv=1&rvrmw=1&rvrmx=1&rvrmy=1&rvrmz=1&rvrna=1&rvrnb=1&rvrnc=1&rvrnd=1&rvrne=1&rvrnf=1&rvrng=1&rvrnh=1&rvrni=1&rvrnj=1&rvrnk=1&rvrnl=1&rvrnm=1&rvrnn=1&rvrno=1&rvrnp=1&rvrnq=1&rvrnr=1&rvrns=1&rvrnt=1&rvrnu=1&rvrnv=1&rvrnw=1&rvrnx=1&rvrny=1&rvrnz=1&rvroa=1&rvrob=1&rvroc=1&rvrod=1&rvroe=1&rvrof=1&rvrog=1&rvroh=1&rvroi=1&rvroj=1&rvrok=1&rvrol=1&rvrom=1&rvron=1&rvroo=1&rvrop=1&rvroq=1&rvror=1&rvros=1&rvrot=1&rvrou=1&rvrov=1&rvrow=1&rvrox=1&rvroy=1&rvroz=1&rvrpa=1&rvrpb=1&rvrpc=1&rvrpd=1&rvrpe=1&rvrpf=1&rvrpg=1&rvrph=1&rvrpi=1&rvrpj=1&rvrpk=1&rvrpl=1&rvrpm=1&rvrpn=1&rvrpo=1&rvrpp=1&rvrpq=1&rvrpr=1&rvrps=1&rvrpt=1&rvrpu=1&rvrpv=1&rvrpw=1&rvrpx=1&rvrpy=1&rvrpz=1&rvrqa=1&rvrqb=1&rvrqc=1&rvrqd=1&rvrqe=1&rvrqf=1&rvrqg=1&rvrqh=1&rvrqi=1&rvrqj=1&rvrqk=1&rvrql=1&rvrqm=1&rvrqn=1&rvrqo=1&rvrqp=1&rvrqq=1&rvrqr=1&rvrqs=1&rvrqt=1&rvrqu=1&rvrqv=1&rvrqw=1&rvrqx=1&rvrqy=1&rvrqz=1&rvrra=1&rvrrb=1&rvrrc=1&rvrrd=1&rvrre=1&rvrrf=1&rvrrg=1&rvrrh=1&rvrri=1&rvrrj=1&rvrrk=1&rvrrl=1&rvrrm=1&rvrrn=1&rvrro=1&rvrrp=1&rvrrq=1&rvrrr=1&rvrrs=1&rvrrt=1&rvrru=1&rvrrv=1&rvrrw=1&rvrrx=1&rvrry=1&rvrrz=1&rvrsa=1&rvrsb=1&rvrsc=1&rvrsd=1&rvrse=1&rvrsf=1&rvrsg=1&rvrsh=1&rvrsi=1&rvrsj=1&rvrsk=1&rvrsl=1&rvrsm=1&rvrsn=1&rvrso=1&rvrsp=1&rvrsq=1&rvrsr=1&rvrss=1&rvrst=1&rvrsu=1&rvrsv=1&rvrsw=1&rvrsx=1&rvrsy=1&rvrsz=1&rvrta=1&rvrtb=1&rvrtc=1&rvrtd=1&rvrte=1&rvrtf=1&rvrtg=1&rvrth=1&rv

Method and Process used for the Case study

The investigation started by opening the working copy of the nitroba_working.pcap file in Wireshark and TShark.

First, I filtered for HTTP GET requests to www.willselfdestruct.com to locate the web activity related to the reported message.

Next, I identified the client IP address 192.168.15.4 and the service IP address 69.25.94.22, then inspected the relevant packets and their timestamps.

I then used the HTTP POST filter to locate packet 83601, which showed the client sending a POST request to /secure/submit, and inspected the packet details and message-related content.

Finally, I followed the relevant TCP stream and reviewed the surrounding packets, including packet 83604 (server response) and packet 83614 (follow-up GET to /secure/success), and also examined the client MAC address and HTTP cookie evidence to support the device-level analysis.

Timeline Report

Time	Packet	Source	Destination	Event
02:04:24.311700	83601	192.168.15.4	69.25.94.22	HTTP POST to /secure/submit
02:04:24.469169	83604	69.25.94.22	192.168.15.4	HTTP response to /secure/submit
02:04:24.564165	83614	192.168.15.4	69.25.94.22	Follow-up HTTP GET to /secure/success

Below is the analysis and result to the eight questions asked on the case study and also relating to all the screenshot above

- **What file did you acquire, and how did you preserve its integrity?**

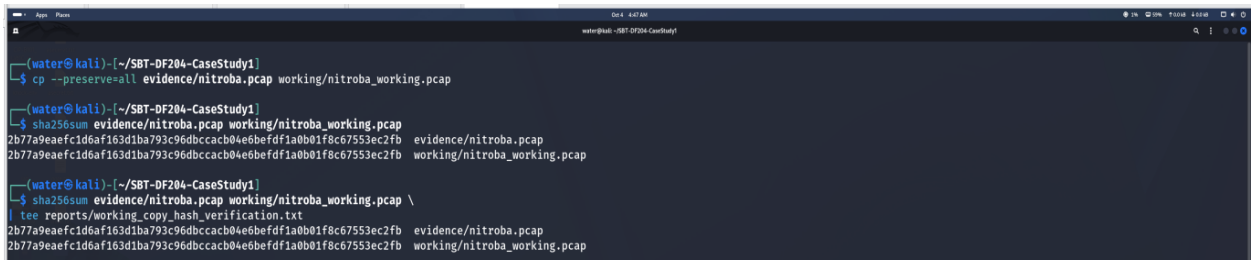
The acquired evidence file was **nitroba.pcap**, obtained from the Digital Corpora case-study traffic capture source.

The file was saved as evidence/nitroba.pcap in the SBT-DF204-CaseStudy1 evidence folder.

The SHA-256 hash was calculated using sha256sum to record and verify the integrity of the original evidence file.

A forensic working copy was created using cp --preserve=all evidence/nitroba.pcap working/nitroba_working.pcap.

The SHA-256 values of the original and working copy were compared and confirmed to be the same, showing that the working copy had not been altered.



2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb evidence/nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb working/nitroba_working.pcap

- **Which client system contacted the web service?**

The client system that contacted the web service was 192.168.15.4.The service IP address was 69.25.94.22, and the traffic was identified as HTTP.The relevant Wireshark filter was ip.src == 192.168.15.4 && ip.dst == 69.25.94.22.The case study also used the HTTP POST filter ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 && http.request.method == "POST".
The relevant POST activity occurred at packet 83601, where the client sent an HTTP POST to /secure/submit at 02:04:24.311700.

Item	**Finding**
Client IP	192.168.15.4
Service IP	69.25.94.22
Protocol	HTTP
Packet number	83601
Packet event	HTTP POST to /secure/submit
Timestamp	02:04:24.311700
Wireshark filter	ip.src == 192.168.15.4 && ip.dst == 69.25.94.22
POST filter	ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 && http.request.method == "POST"

- **Evidence Linking the Client to the Harassment Message**

The evidence shows that client IP 192.168.15.4 sent an HTTP POST request to the web service at 69.25.94.22. The relevant request was packet 83601, at 02:04:24.311700, using the URI /secure/submit. The investigation used the filter:
ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 && http.request.method == "POST"
The Wireshark analysis also searched for the message content using data-text-lines contains "secure anonymous E-mail", linking the HTTP traffic to the reported harassment-message activity. The POST request was followed by an HTTP response and then a GET request to /secure/success, supporting that the submission was processed by the web service

Evidence	**Result**
Client IP	192.168.15.4
Service IP	69.25.94.22
HTTP method	POST
Packet	83601
Time	02:04:24.311700
URI	/secure/submit
Message search	data-text-lines contains "secure anonymous E-mail"
Follow-up	GET /secure/success

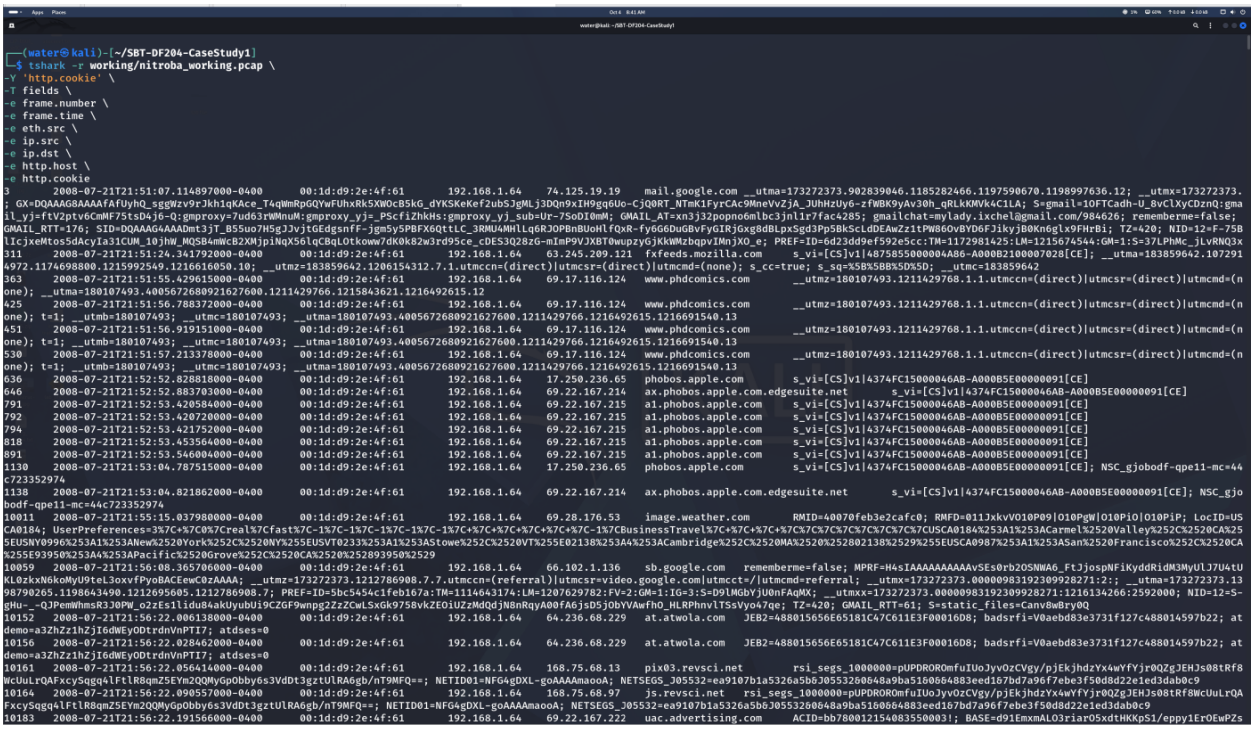
- **Device That Made the Relevant Request**

The device that made the relevant request had the client IP address 192.168.15.4 and the MAC address 00:17:f2:e2:c0. The MAC address was identified using the Wireshark/TShark filter eth.addr == 00:17:f2:e2:c0:ce and was associated with the client traffic.
The relevant client traffic was also checked using eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@".

- **Who Is Associated With the Device?**

The cookie/browser evidence associates the relevant device with the Gmail address jcoach@gmail.com. In packet 78990, the traffic came from client IP 192.168.15.4 and MAC

address 00:17:f2:e2:c0, the same IP/MAC pair identified during the harassment-message investigation. The HTTP cookie in this packet revealed the email address jcoach@gmail.com, providing the scenario-supported link between the device and the account.
The email address corresponds to Johnny Coach, who is listed on the Chemistry 109 class roster. Therefore, the evidence supports an association between the device and Johnny Coach's Gmail account, but this should not be stated as absolute proof that Johnny personally sent the message because the device was operating through a shared/open wireless network.
Packet evidence: Packet 78990
Client IP: 192.168.15.4
MAC address: 00:17:f2:e2:c0:ce
Cookie identity: jcoach@gmail.com
Associated person: Johnny Coach



- **Is That Person on the Chem 109 Roster?**

Yes. The supported identity from the browser/cookie evidence shows Johnny Coach, which is also associated with the email address jcoach@gmail.com. The provided Chemistry 109 roster contains Johnny Coach, so the identity supported by the cookie evidence matches a person on the roster. The basis for this comparison is the jcoach@gmail.com account found in the client traffic and the matching Johnny Coach entry in the supplied Chem 109 roster.

This establishes that the supported account identity is associated with a Chem 109 student. However, it does not by itself prove that Johnny Coach personally sent the harassment message, because the investigation involves a shared/open wireless network.

- **When Did the Activity Occur?**

The activity occurred on 22 July 2008, according to the exact timestamps recorded in the PCAP. The timestamps are displayed with the -0400 time-zone offset (UTC−4). Converting these times to West Africa Time (WAT, UTC+1) adds 5 hours.

PCAP Time (UTC−4)	Packet	Event	WAT (UTC+1)
02:04:24.311700	83601	HTTP POST to /secure/submit	07:04:24.311700
02:04:24.469169	83604	HTTP response to /secure/submit	07:04:24.469169
02:04:24.564165	83614	Follow-up HTTP GET to /secure/success	07:04:24.564165

The PCAP evidence confirms packet 83601 as the relevant submission, followed approximately 0.157 seconds later by the server response and another 0.095 seconds later by the successful follow-up request. The detailed PCAP output also records the full date and -0400 offset for all three packets.

Therefore, the relevant web submission occurred at approximately 02:04:24 on 22 July 2008 in the PCAP's displayed UTC−4 time, or 07:04:24 WAT.

- **What Conclusion Can You Defend?**

The PCAP provides strong evidence that the client device with MAC address 00:17:f2:e2:c0 and IP address 192.168.15.4 submitted the relevant message to willselfdestruct.com. The HTTP POST in packet 83601 was sent to 69.25.94.22 at 02:04:24.311700 (UTC−4), followed by a successful request to /secure/success. The cookie/browser evidence associates the device with jcoach@gmail.com, which corresponds to Johnny Coach on the Chem 109 roster. Therefore, the evidence supports a moderate-to-high confidence association between the device, the account, and the reported activity, but it does not provide absolute proof that Johnny Coach personally sent the message.

Limitations and Alternative Explanations

1. Shared/open wireless network: Other people could have used the same network, so the client IP address alone cannot identify the individual who sent the message.
2. Device/account use: The cookie links the device to the jcoach@gmail.com account, but another person could potentially have used the device or account.
3. MAC address limitation: The MAC address identifies the network interface involved in the traffic, not necessarily the person operating the device.

4. Attribution limitation: The PCAP demonstrates network activity and an account association, but it does not independently prove who physically initiated the request.

Overall conclusion: The evidence supports that the identified device and associated jcoach@gmail.com account were involved in the relevant web activity, with moderate-to-high confidence. However, the evidence should not be stated as conclusive proof that Johnny Coach personally authored or sent the harassment message.

- **Evidence Log**

Evidence ID	**Packet or item**	**Finding**	**Why it matters**	**Screenshot or appendix**
E01	Original nitroba.pcap and working copy nitroba_working.pcap	The original PCAP was preserved and copied to the working folder using cp --preserve=all. The SHA-256 values of the original and working copy were confirmed to be the same.	This shows that the evidence was preserved and that the analysis was performed on a matching working copy without changing the original evidence.	Evidence Integrity / SHA-256 Screenshot
E02	HTTP traffic involving www.willselfdestruct.com	The PCAP shows web activity from client IP 192.168.15.4 communicating with service IP 69.25.94.22.	This identifies the client system and web service involved in the relevant activity and provides the network connection used for further analysis.	Client and Service IP Screenshot
E03	Packet 83601, HTTP POST	At 02:04:24.311700 -0400, client 192.168.15.4 sent an HTTP POST to 69.25.94.22, host www.willselfdestruct.com, URI /secure/submit. The POST was correlated with the search for the reported	This is the strongest packet-level evidence connecting the client traffic to the reported harassment-message activity.	HTTP POST / Packet 83601 Screenshot

		message content, including secure anonymous E-mail.		
E04	Ethernet/MAC information for client traffic	The client IP 192.168.15.4 is associated with MAC address 00:17:f2:e2:c0 in the analysed traffic.	The MAC address provides a more specific device-level identifier than the IP address. However, it still does not by itself prove which person was using the device.	Client MAC Address Screenshot
E05	HTTP cookie/browser evidence	Cookie/browser evidence was examined using the client MAC address and HTTP cookie filters. This evidence was used to assess whether the device could be linked to an identity supplied in the case materials.	Cookie or browser evidence can provide an additional link between the device and a scenario-provided identity, but the link must be based only on the evidence actually present in the case materials.	Cookie/Browser Evidence Screenshot
E06	Timeline: Packets 83601, 83604 and 83614	Packet 83601 shows the POST at 02:04:24.311700, packet 83604 shows the server response at 02:04:24.469169, and packet 83614 shows a follow-up GET to /secure/success at 02:04:24.564165.	The sequence confirms that the POST, server response and successful follow-up request occurred as one closely connected web transaction. It strengthens the correlation between the client activity and the reported message submission.	Timeline / HTTP Transaction Screenshot

2.7. Analysis and Findings

- The investigation identified 192.168.15.4 as the client IP and 69.25.94.22 as the web-service IP involved in the relevant activity.
- Wireshark and TShark identified packet 83601 at 02:04:24.311700 -0400, showing an HTTP POST to /secure/submit on www.willselfdestruct.com.
- The server responded in packet 83604, followed by a GET request to /secure/success in packet 83614, supporting the sequence of the web activity.
- The client device was associated with MAC address 00:17:f2:e2:c0, and available cookie/browser evidence was examined for an identity link.
- The evidence supports that the client device accessed the service and submitted the relevant request, but the shared open wireless network means the traffic alone cannot prove which person used the device.

2.8. Challenges and Solutions

One challenge was identifying the exact web traffic because the PCAP contained many different network connections, this was solved by using Wireshark and Tshark filters for www.willselfdestruct.com, HTTP POST requests, the client IP, and the service IP.
Another challenge was connecting the network activity to a particular device and person without making unsupported assumptions. The open shared wireless network remained a limitation because another person using the same network could potentially generate similar traffic.

2.9. Conclusion

- The investigation found clear evidence that 192.168.15.4 communicated with 69.25.94.22 and accessed the willselfdestruct.com service.
- Packet 83601 provides direct evidence of the relevant HTTP POST to /secure/submit, followed by the server response and success request.
- The traffic can reasonably be linked to the identified client device with MAC address 00:17:f2:e2:c0.

However, the evidence does not by itself prove that a specific Chemistry 109 student personally sent the message because the network was shared and used an open wireless router.

2.10. Recommendations

Assignment like this improved student knowledge and understanding on Forensic case giving the student the ability to think deep and do deep research before concluding on a case

2.11. References

Class Video Guide from Founder ICDFA on Basic Networking Skills for Digital Forensics
Class assignments from Founder ICDFA on Basic Networking Skills for Digital Forensics
Forensic-Analysis Report guide from https://www.cliffsnotes.com/study-notes/16611977
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